VAJIRAM & RAVI

Institute for IAS Examination

PowerUp

Current Affairs Test Series – 2025

CA Test – 14 – (CA2264)

Answer Key &
Detailed Answer Explanations

CA TEST – 14 - Answer Key									
1. (c)	11. (b)	21. (a)	31. (b)	41. (c)	51. (b)	61. (b)	71. (c)	81. (a)	91. (a)
2. (d)	12. (b)	22. (b)	32. (c)	42. (a)	52. (c)	62. (a)	72. (b)	82. (b)	92. (a)
3. (a)	13. (c)	23. (b)	33. (a)	43. (c)	53. (c)	63. (a)	73. (b)	83. (c)	93. (d)
4. (d)	14. (c)	24. (c)	34. (c)	44. (b)	54. (c)	64. (a)	74. (d)	84. (c)	94. (b)
5. (c)	15. (c)	25. (a)	35. (a)	45. (b)	55. (b)	65. (b)	75. (c)	85. (d)	95. (c)
6. (a)	16. (d)	26. (c)	36. (a)	46. (b)	56. (b)	66. (d)	76. (c)	86. (b)	96. (a)
7. (c)	17. (b)	27. (c)	37. (a)	47. (b)	57. (a)	67. (c)	77. (d)	87. (c)	97. (c)
8. (d)	18. (b)	28. (c)	38. (c)	48. (a)	58. (b)	68. (d)	78. (a)	88. (a)	98. (b)
9. (c)	19. (c)	29. (d)	39. (d)	49. (b)	59. (a)	69. (a)	79. (b)	89. (d)	99. (a)
10. (b)	20. (b)	30. (d)	40. (d)	50. (d)	60. (c)	70. (b)	80. (d)	90. (d)	100.(c)

Q1.
Answer: c
Explanation:
Cane Molasses is the mother liquor left over, after production of sugar in a factory . It hass following subtypes:

- A-Heavy molasses: It is an-intermediate by-product produced during production of sugar having purity of molasses between 68 percent to 72 percent.
- **B-Heavy molasses:** It is an-intermediate by-product produced during production of sugar having purity of molasses between **48 percent to 52 percent.**
- C-Heavy molasses: It is a final by-product produced during production of sugar having purity of molasses between 28 percent to 32 percent.
- Ethanol, a 99.9% pure alcohol, is added to fuel to reduce the nation's dependence on imported crude oil. Ethanol in India is primarily produced from sugarcane-based molasses and grain-based sources. Among these, **B-heavy molasses is a major contributor, accounting for 60% of ethanol production**, followed by sugarcane juice at 20%, while C-heavy molasses has a minor role. **So, statements 1 and 2 are correct.**

Therefore, option (c) is the correct answer.
Relevance: Sugar manufacturers' association urges the government to hike ethanol prices after a two-year freeze.

Q2.
Answer: d
Explanation:

- **The Green Credit Programme (GCP)** was dubbed as a mechanism to incentivise "pro-planet" actions in response to climate change, and one which would promote the Centre's Mission LiFE (Lifestyle for Sustainable Environment). **So, statement 1 is not correct.**

- At its core, the GCP seeks voluntary participation (read investments) of individuals, companies, industries, and other entities across seven different activities, such as tree plantation, waste management, and water conservation, that will lead to improvements in the environment.
    - GCP mandates use of degraded forest land for plantations. It also has complementary provisions which allow exchange of credits, generated by paying money, for meeting the compensatory afforestation compliance. **So, statement 2 is correct**.
- **Green Credit generated under the programme can be traded on a domestic market platform** to potential buyers looking to meet sustainability targets or existing legal obligations. For instance, green credits could be exchanged for meeting compliance with compensatory afforestation, in cases where forest land has been used for development projects. The credits can also be used by listed companies as part of their environmental, social, and governance disclosures under SEBI's Business Responsibility and Sustainability framework, indicating efforts taken by these companies on environmental sustainability. **So, statement 3 is correct.**

Therefore, option (d) is the correct answer.

Relevance: The Green Credit Programme (GCP) seeks to invite voluntary participation in sectors ranging from plantations to water conservation in exchange of tradable credits.

Q3.

Answer: a

Explanation:

- The Biomass mission will map the world's forests to provide the very first comprehensive measurements of forest biomass at a global scale. It will find out how forests are changing, and further our understanding about their role in the carbon cycle.
- The mission will allow scientists to more accurately measure how the distribution of carbon in the planet is changing, as humans continue to cut down trees and increase CO2 levels in the atmosphere. **So, statement 1 is correct.**
- **It is an European Space Agency (ESA) satellite.** To fulfil its objectives, the mission will use a synthetic aperture radar (SAR) to map the Earth's surface. Notably, the satellite, which is fitted with a massive 12-m antenna, will be the first in space to use a long-wave P-band SAR. It will thus be able to penetrate dense forest canopies to assess how much carbon is stored on the floor and in the branches of the trees, and to assess how these levels are changing over time**. So, statement 2 is not correct.**

Therefore, option (a) is the correct answer.

Relevance: The European Space Agency (ESA) launched its ambitious Biomass mission on April 29.

Q4.

Answer: d

Explanation:

- The Waqf (Amendment) Bill, 2025, aims to update the Waqf Act, 1995, to fix issues in the management of Waqf properties. In 1995, the Waqf Act allowed Waqf to be formed by declaration, user, or endowment (waqf-alal-aulad). **The 2025 amendment removes waqf by user and allows formation only through declaration or endowment. So, statement 1 is not correct.**
- In 1995, the Waqf Act required that all members of the Central Waqf Council be Muslims, including at least two women members. **The 2025 amendment bill provides that two members must be non-Muslims.** The Bill also empowers the state government to nominate one person from each background to the Waqf Board. They need not be Muslims. **So, statement 3 is correct.**
- The 1995 Waqf Act provides for Assigned survey commissioners and additional commissioners to conduct Waqf surveys. The 2025 amendment empowers **Collectors** to conduct surveys and mandates that pending surveys be **conducted as per state revenue laws. So, statement 2 is correct.**

- **Under the Waqf (Amendment) Bill, 2025, Separate Waqf boards are allowed** for Bohra and Agakhani sects, along with Shia and Sunni sects. **So, statement 4 is correct.**

Therefore, option (d) is the correct answer.
Relevance: The Supreme Court to hear a batch of pleas challenging the Waqf (Amendment) Act's constitutional validity, weeks after the government paused two central aspects of the contentious law.

Q5.
Answer: c
Explanation:

- **Professional tax** is a tax on all kinds of **professions, trades, and employment and is levied based on the income of such professions, trades, and employment**. It is levied on employees, a person carrying on the business, including freelancers, professionals, etc., subject to income exceeding the monetary threshold, if any.
- An individual can get a tax **exemption under the Income-tax Act, 1961** on the professional tax paid.
- However, each of the state governments levying professional tax has drafted their own law. Article 276 of the Constitution empowers States to levy taxes on professions, trades, callings, and employments.
- The applicability of professional tax, the tax rate based on the slabs and other aspects of professional tax differs from state to state.

Therefore, option (c) is the correct answer.
Relevance: Maharashtra plans to transfer professional tax collection rights to local bodies.

Q6.
Answer: a
Explanation:

- The IPU is the global organization of national parliaments. It began in 1889 as a small group of parliamentarians dedicated to promoting peace through parliamentary diplomacy and dialogue, and has since grown into a truly global organization with 181 Members and 14 Associate Members. The IPU facilitates parliamentary diplomacy and empowers parliaments and parliamentarians to promote peace, democracy, and sustainable development around the world. **So, statement 1 is correct.**
- **IPU is financed primarily by member states' public funds, with its headquarters in Geneva, Switzerland.** It also has offices in New York, United States of America, and Vienna, Austria. **So, statement 2 is correct and statement 3 is not correct.**

Therefore, option (a) is the correct answer.
Relevance: The IPU's next Assembly is hosted by Uzbekistan, which has invited all member parliaments of the IPU to the 150th Assembly in Tashkent.

Q7.
Answer: c
Explanation:

- **"Kannadippaya" (meaning 'mirror mat')** originates from its distinct reflective design. It is **woven using reed bamboo's soft inner layers**, which give it remarkable thermal properties—it provides warmth in winter and a cooling effect in summer.

- It is **primarily crafted by tribal communities** such as the Oorali, Mannan, Muthuva, Malayan, and Kadar tribes, along with Ulladan, Malayarayan, and Hill Pulaya artisans in the Idukki, Thrissur, Ernakulam, and Palakkad districts.
- The finest Kannadippaya is ***made from Teinostachyum wightii***, locally known by various names such as Njoonjileetta, Njoojoora, Ponneetta, Meieeta, and Neytheetta.
- Other bamboo species, like Ochlandra sp. (locally called Kareetta, Pereetta, Velleeta, Chitoora, and Kanjoora), are also used.
- Kannadippaya is the **first tribal handicraft from Kerala to receive a GI tag.**

Therefore, option (c) is the correct answer.
Relevance: Kannadippaya has received the Geographical Indication (GI) tag, ensuring market protection and a global platform for this traditional product.

Q8.
Answer: d
Explanation:

- **Organs-on-chips (OoCs) are systems containing engineered or natural miniature tissues grown inside microfluidic chips**. To better mimic human physiology, the chips are designed to control cell microenvironments and maintain tissue-specific functions. **So, statement 1 is correct.**
- Drug development is notoriously slow and expensive because of the traditional reliance on testing drugs in animals before they are tested in humans. **OoC overcomes this limitation by providing living, three-dimensional cross-sections of human organs. So, statement 2 is correct.**
- **Given the limitations associated with animal testing**, in recent years, there has been interest in developing faster, less expensive, and more informative new approaches to gathering toxicological information. These methods, which researchers aspire to develop for use in place of traditional animal testing, are often referred to as new approach methods (NAMs**). The NAMs include 3D spheroids, organoids, bioprinting, and organ-on-chips. So, statement 3 is correct.**

Therefore, option (d) is the correct answer.

Q9.
Answer: c
Explanation:

- **India Volatility Index, is a crucial metric calculated by the National Stock Exchange (NSE) to measure the market's anticipation of volatility and fluctuations in the near term**. Introduced in 2003 by the NSE, it provides insights into market volatility over the next 30 days. Unlike price indices like NIFTY, which track the market's direction based on equity price movements, India VIX focuses solely on volatility. It derives its value using the Black-Scholes model, considering variables like strike price, market price of the stock, time to expiry, risk-free rate, and volatility. Essentially, it helps investors understand the market's volatility before making investment decisions.
- The India nifty VIX is **a reliable indicator for investors across various segments.** For intraday traders, it informs about market volatility, aiding in risk assessment and determining stop-loss levels. Long-term investors utilise it to get through potential losses and make informed investment decisions. Options traders find it invaluable in deciding whether to buy or sell options based on market volatility. Portfolio and mutual fund managers use its insights to optimise their investment strategies, choosing between high and low-beta stocks.

Therefore, option (c) is the correct answer.
Relevance: Volatility index rises 6.5% amid growing India-Pak tension.

Q10.
Answer: b
Explanation:

- The anti-defection law sought to prevent political defections, which may be due to the reward of office or other similar considerations. The law initially stated that the decision of the Presiding Officer is not subject to judicial review**. This condition was struck down by the Supreme Court in 1992, thereby allowing appeals against the Presiding Officer's decision in the High Court and the Supreme Court. So, point 1 is correct.**
- The law does not specify a time period for the Presiding Officer to decide on a disqualification plea. Given that courts can intervene only after the Presiding Officer has decided on the matter, the petitioner seeking disqualification has no option but to wait for this decision to be made**. Until now, the apex court had treaded carefully in its judgments, urging Speakers to decide anti-defection pleas within a "reasonable time". The court had not specified what the "reasonable" time ought to be in due regard to the Constitutional office of the Speaker. So, point 2 is not correct.**
- The law provides for a member to be disqualified if he 'voluntarily gives up his membership'. **However, the Supreme Court has interpreted that in the absence of a formal resignation by the member, the giving up of membership can be inferred by his conduct. In other judgments, members who have publicly expressed opposition to their party or support for another party were deemed to have resigned. So, point 3 is correct.**

So, only two of the above are correct with respect to the role of the Supreme Court in anti-defection law.
Therefore, option (b) is the correct answer.
Relevance: Supreme court while hearing petitions seeking timely action by the Telangana Assembly Speaker on disqualification proceedings pending against 10 MLAs asked if constitutional courts cannot direct Speakers.

Q11.
Answer: b
Explanation:
The Gulf of Mannar Marine Biosphere Reserve, located along the southeast coast of Tamil Nadu, is one of the most biologically diverse marine areas in mainland India. It is recognized for its rich marine life, including **coral reefs, seagrasses, and endangered species like the Dugong and sea turtles.**

- The Gulf of Mannar is often referred to as one due to its extraordinary biodiversity, with over 4,223 species of flora and fauna. It includes coral reefs, seagrass meadows, mangroves, and marine animals, making it ecologically significant.
- While the reserve is a critical habitat for the critically endangered Dugong (sea cow), it has not been officially declared as India's first "Conservation Reserve" for Dugong. The country's first **'Dugong Conservation Reserve' is in Palk Bay covering the coastal waters of Thanjavur and Pudukottai districts**. **So, statement 1 is not correct.**
- The Gulf of Mannar was declared as a Marine National Park in 1986 and later designated as **India's first Marine Biosphere Reserve in 1989 under UNESCO's Man and the Biosphere Programme**. It holds the distinction of being the first marine biosphere reserve in India and in South & Southeast Asia. **So, statement 2 is correct.**

Knowledge Box
Gulf of Mannar: ● **Location:** Tamil Nadu, between Rameswaram and Kanyakumari ● **Area Covered:** ~10,500 sq. km including 21 islands (2 submerged)

- **Key Marine Habitats:** Coral reefs, seagrass beds, mangroves
- **Important Fauna:** Dugong, green and hawksbill turtles, whale shark, sea horses
- **Governance**: Managed by the Gulf of Mannar Biosphere Reserve Trust (GOMBRT), established by the Tamil Nadu government in 2000

Therefore, option (b) is the correct answer.

Q12.
Answer: b
Explanation:

- The Third Launch Pad (TLP) will be established at Sriharikota. The project has been approved by the Union Cabinet and financial sanction has been obtained for a total budget outlay of ₹3984.86 Crore. Establishment of the pad is envisaged to be completed within 4 years timeframe.
- ISRO's **Next Generation Launch Vehicle (NGLV),** which is under development is about 90 m tall with a maximum lift-off mass of approximately 1000 tonne. Existing launch pads at Sriharikota cannot launch this class of vehicles. The propellant servicing facilities and the Umbilical Tower of the existing launch pads are not designed to meet the requirements of the new propulsion system based on Liquid Methane.
- In view of very large height & size, the next generation of launch vehicles are planned with **horizontal integration and transport, which are then tilted onto the launch pad along with a Tiltable Umbilical Tower (TUT)**. Also, TLP incorporates necessary features in terms of foundation support & servicing requirements for future augmentation towards supporting the launches of India's Crewed Lunar mission.
- The first stage of NGLV is configured with a cluster of 9 engines. The hot testing of this stage is planned at the Launch Pad, thereby eliminating the need for establishing a huge separate facility for stage testing.

Therefore, option (b) is the correct answer.
Relevance: Third Launch Pad (TLP) will be established at Sriharikota.

Q13.
Answer: c
Explanation:
The **rare earth elements (REEs) are 17 metals in Group 3 of the Periodic Table** comprising **Lanthanide series elements and Scandium and Yttrium** (due to similar physical properties and found in the same ores and deposits). Contrary to their name, they are **moderately abundant in nature** but not concentrated enough to make them economically exploitable. **India** has the **fifth-largest rare earth resource globally,** with mainly concentrated in its monazite minerals.

- **Monazite** is a **phosphate mineral rich in light REEs like cerium, lanthanum, and neodymium.** It is commonly found in **placer deposits and beach sands**, particularly along the Indian coasts. Due to its **thorium** content, it is also significant for its nuclear potential. **So, point 1 is correct.**
- **Bauxite** is primarily an aluminium ore. While it may contain trace amounts of REEs, it is not a commercially viable source for rare earth extraction. Hence, it is not considered a major REE mineral. **So, point 2 is not correct.**
- **Bastnaesite** is a **carbonate-fluoride mineral and a leading source of light REEs such as cerium and lanthanum.** It occurs in igneous rocks and is extensively mined in the USA and China. Its composition makes it ideal for extracting REEs used in magnets and catalysts. **So, point 3 is correct.**

- **Xenotime** is a **phosphate** mineral that contains heavy REEs, **particularly yttrium and dysprosium.** It is typically found in **granite pegmatites and metamorphic rocks.** Xenotime is crucial for advanced applications, including lasers and phosphors. **So, point 4 is correct.**

So, only three of the above are major sources of Rare Earth Elements (REEs).
Relevance: In order to reduce the import dependency of Rare Earth Elements (REE), Atomic Minerals Directorate for Exploration and Research (AMD) is carrying out exploration to augment resources of REE.
Therefore, option (c) is the correct answer.

Q14.
Answer: c
Explanation:

- **Domestic prices of perishable agriculture produces** are mainly influenced by factors such as demand and supply, trade policies, effective taxes and duties, etc. The Central Government takes necessary measures as and when required to balance the demand and supply scenario of agriculture and horticulture produces in the domestic market and ensure remunerative prices to the farmers through appropriate policy measures and market intervention scheme.
- To safeguard farmers from market volatilities, the Government takes various measures through different schemes and programmes. These include **Market Intervention Scheme (MIS) for perishable horticultural and agricultural commodities**; Integrated Scheme for Agriculture Marketing (ISAM) to promote- creation of scientific storage capacity; National Agriculture Market (e-NAM) online trading platform for better price discovery through competitive online bidding etc.
- Nationwide information network system for reporting and dissemination of market price information on agriculture commodity is provided through Agmarknet web portal. Under Agriculture Infrastructure Fund (AIF), Govt. provides a medium-long term loan facility for investment in viable projects for post-harvest market infrastructure including warehousing facility and community farming assets through interest subvention and financial support.
- Government has introduced a **new component of Price Differential Payment (PDP) under Market intervention scheme (MIS) from 2024-25** season for direct payment of the price **difference between the Market Intervention Price (MIP) and the selling price to the farmers of perishable crops**. States/UTs have an option to choose either to do physical procurement of the crop or to make the differential payment between the MIP & Sale Price to the farmers. Further, from 2024-25 season, Government added another component under Market intervention scheme for **reimbursing the Storage and Transportation cost of TOP crops (Tomato, Onion and Potato)** to central nodal agencies for transporting them from the producing state to consuming states in the interest of the farmers.
- To provide remunerative price to the farmers, Ministry of Agriculture & Farmers Welfare implements Market Intervention Scheme (MIS), **a component under Pradhan Mantri Annadata Aay Sanrakshan Abhiyan (PM-AASHA)**, for procurement of agricultural and horticultural commodities which are perishable in nature and are not covered under the Price Support Scheme (PSS). The objective of intervention is to **protect the growers of these commodities from making distress sale in the event of a bumper crop** during the peak arrival period when the prices tend to fall below economic levels and the cost of production. The scheme is implemented at the request of a State/UT government, which is ready to bear 50 percent of the loss (25 percent in case of North-Eastern States), if any, incurred on its implementation.

Therefore, option (c) is the correct answer.

Q15.
Answer: c
Explanation:

- **Chlorpyrifos**, a pesticide classified as 'moderately hazardous' by the World Health Organisation, is still **approved for use in India on 18 crops, despite bans in over 40 countries.** It is linked to neurotoxicity, reproductive toxicity, irreversible brain damage in unborn children, and the ability to contaminate distant ecosystems due to its mobility. **So, statement 1 is correct and statement 2 is not correct.**
- **Pesticide Action Network (PAN) India:**
    - It advocates placing chlorpyrifos under **Annex III of the Rotterdam Convention**, requiring prior informed consent before trade. **So, statement 3 is correct.**
    - It calls for its inclusion under Annex A of the Stockholm Convention, which demands a complete global ban, though exemptions are usually allowed.
    - It stresses that safer alternatives are already available, making a total ban both feasible and necessary.
- Unauthorised Use in India: A 2022 report found the illegal use of chlorpyrifos and other agrochemicals like paraquat, raising serious regulatory and enforcement concerns.

Therefore, option (c) is the correct answer.

Q16.
Answer: d
Explanation:

- The **Kerala Urban Policy Commission (KUPC),** first state level urban commission in India, was constituted in December 2023 to frame an urban policy.
- KUPC stated in a report that by 2050, 80 percent of the state's population will be urban. The rate of urbanisation in Kerala is higher than the national average.
- As cited in the latest Economic Survey 2023-24, it is expected that by 2030, more than 40 percent of India's population will live in urban areas. This estimation has been made on the basis of studies and reports of NITI Aayog.
- **Kerala is now the only Indian state to have an urban policy.** This is a significant milestone in Kerala's development history because the report will provide insights for establishing development policies for the state over the next 25 years, particularly in light of rising urbanisation and climate change.
- The commission suggested introducing a **targeted green fee** on urban development projects to fund disaster risk reduction initiatives, including **resilient infrastructure and emergency response systems**. It also called for expanding disaster-specific insurance schemes to protect vulnerable communities and critical assets.
- A state-level climate finance advisory cell should be established to assist local governments in **accessing green funds**, **carbon financing**, and **international climate adaptation resources**, ensuring efficient allocation for mitigation and resilience-building efforts.

Therefore, option (d) is the correct answer.
Relevance: Kerala has recently become the first state in India to establish an Urban Commission.

Q17.
Answer: b
Explanation:
Hormones are the body's messengers, and they play an important role in sustaining a variety of physiological functions, including stress reactions and energy control. Among these, oxytocin, cortisol, and insulin are particularly important for both physical and emotional well-being.

- **Oxytocin** is commonly associated with pleasant emotions such as trust, bonding, and social connection. It improves appetite regulation and emotional well-being. **So, statement 1 is correct.**
- **Cortisol**, the "stress hormone," is released under stressful conditions to assist the body in responding properly. High cortisol levels, which are frequently caused by prolonged stress, can result in weight gain, mood changes, and impaired metabolism. **So, statement 2 is not correct.**
- **Insulin** helps regulate blood sugar levels by promoting glucose uptake into cells for energy. Stable insulin levels are critical for avoiding energy dumps and controlling cravings. **So, statement 3 is not correct.**

Therefore, option (b) is the answer.

Q18.
Answer: b
Explanation:

- **SpaceX** is preparing to launch a crew of four private astronauts led by a cryptocurrency entrepreneur on a trip to orbit Earth from pole to pole, a new track that no humans have taken before. **So, statement 1 is not correct.**
- The Fram2 mission will place four humans in polar orbit, **also known as low Earth orbit, for the first time in human history.** The international crew will be placed in a 90-degree circular orbit, allowing them to fly over both the North and South Poles. During their voyage, they are anticipated to see natural phenomena including the Aurora Borealis. **So, statement 2 is correct.**
- Fram2 will be SpaceX's 16th crewed trip using the reusable Crew Dragon, a gumdrop-shaped spacecraft designed with NASA funding to transport people to and from the International Space Station.
- Fram2 will be a multi-day space mission in which the Dragon spacecraft and crew will tour Earth from polar orbit, travelling over the polar regions.
- The astronauts will undertake 22 research studies to better understand human health in orbit. Their studies include the first X-ray in space to **analyse muscle and skeletal mass, as well as the growth of mushrooms in microgravity.**

Therefore, option (b) is the correct answer.
Relevance: Elon Musk's SpaceX is preparing to launch a crew of four private astronauts led by a cryptocurrency entrepreneur on a trip to orbit Earth from pole to pole, a new track that no humans have taken before.

Q19.
Answer: c
Explanation:

- The government has notified draft Rules introducing targets for the reduction of greenhouse gas (GHG) emissions by "obligated entities" in energy-intensive sectors and industries. The Draft Greenhouse Gases Emissions Intensity (GEI) Target Rules, 2025, notified by the Ministry of Environment, Forest and Climate Change on April 16, puts in place a compliance mechanism for the Carbon Credit Trading Scheme, 2023 (CCTS).

- **Greenhouse Gases (GHGs)** are gases that trap heat in the atmosphere and contribute to the "greenhouse effect" that raises surface temperature on Earth.
    - The five most abundant GHGs in the atmosphere are water vapour, carbon dioxide, methane, nitrous oxide and ozone. Other GHGs include synthetic fluorinated gases such as **chlorofluorocarbons (CFCs) and hydrochlorofluorocarbons (HCFCs).**
- ***Greenhouse Gases Emissions Intensity*** or GEI refers to **the amount of GHGs that are emitted per unit of product output.** It is the amount of GHGs that are released in the production of, say, 1 tonne of cement, aluminium, or paper, etc.
- The draft Rules define GEI as "greenhouse gases emission intensity in **tCO2e/** equivalent output or product". tCO2e, or **tonnes of carbon dioxide equivalent**, is the standard unit used to measure the impact of all GHGs, not just CO2, based on their potential to warm the planet.
- Carbon Emissions Intensity refers to the amount of carbon dioxide (CO2) emitted per unit of economic output or activity.

Therefore, option (c) is the correct answer.
Relevance: Recently, the government of India has notified draft Rules introducing targets for the reduction of greenhouse gas (GHG) emissions.

Q20.
Answer: b
Explanation:

- The **Ministry of Women and Child Development** launched the SHe-Box Portal. The Sexual Harassment Electronic Box (SHe-Box) is a Government of India program that provides a single-window platform for women to report workplace sexual harassment. **So, statement 1 is not correct.**
- It is available to all women, **regardless of their employment status** (organised or unorganised, public or private). **So, statement 2 is correct.**
- When a complaint is filed on SHe-Box, it is automatically forwarded to the proper authority for further action. This tool enables prompt resolution and responsibility for workplace harassment incidents.
- Portal can generate monitoring reports by Ministry, Department, State, UT, Private Sector, or District.
- Built as per the SH Act, 2013 and inquiry must be completed within 90 days as per the Act.

Therefore, option (b) is the correct answer.
Relevance: The Ministry of Women and Child Development recently launched the SHe-Box portal to help in better implementation of various provisions of 'The Sexual Harassment of Women at Workplace Act, 2013' (SH Act).

Q21.
Answer: a
Explanation:

- The Joint Polar Satellite System (JPSS) is **an advanced weather satellite system** of **National Oceanic and Atmospheric Administration (NOAA).** JPSS is a collaborative program between the National Oceanic and Atmospheric Administration (NOAA) and its acquisition agent, National Aeronautics and Space Administration (NASA).

- **JPSS constellation consist of five satellites**, with the first three currently in orbit. JPSS-1 and JPSS-2, which were renamed NOAA-20 and NOAA-21, are operational with the fleet's pathfinder satellite, the Suomi National Polar-orbiting Partnership (S-NPP). JPSS-4 will be the next satellite to launch in the series, followed by JPSS-3, the final satellite in the series to launch.

- JPSS helps scientists across the world study Earth. The data helps scientists forecast severe weather events such as blizzards, hurricanes, and tornadoes.
- Joint Polar Satellite System (JPSS) provides **observations that serve as the backbone of both short- and long-term weather forecasts,** including those that help us predict and prepare for severe weather events. **So, statement 1 is correct.**
- JPSS satellites orbit Earth from pole to pole **14 times a day**, ensuring full global coverage twice daily. **So, statement 3 is correct.**
- JPSS collects data about the Earth by measuring **microwave, infrared, visible, and ultraviolet light** and uses radio waves to send weather data to ground stations near the North and South Poles. **So, statement 3 is not correct.**

Therefore, option (a) is the correct answer.

Q22.
Answer: b
Explanation:

- **Sir Syed Ahmad Khan,** who founded the Aligarh Movement in 1875, was a 19th-century Indian Muslim scholar and reformer. Born in a Muslim family, he served in the British judicial system. After retiring in 1876, he joined the Imperial Legislative Council in 1878, earning a knighthood in 1888 for his loyalty.
- Sir Syed sought to blend Western education with Islamic teachings, advocating for a rational, adaptable interpretation of the Quran and religion.
- He emphasized that religious principles must evolve to stay relevant. As an educationist, Sir Syed opened schools, translated books into Urdu and founded the Mohammedan Anglo-Oriental College in Aligarh in 1875.
- **Sir Syed Ahmad Khan** launched three significant bilingual periodicals — **The Loyal Muhammadans of India (1860), The Aligarh Institute Gazette (1866) and Tahzib-ul-Akhlaq (1870)** — to promote modernity, rational thought, and cultural pluralism, while countering sectarianism and bigotry. **So, points 1, 2 and 3 are correct.**
    - The **Loyal Mohammedans of India** was a newspaper and series of pamphlets aimed at highlighting Muslim loyalty to the British during the 1857 revolt and improving their image in British India.
    - The **Aligarh Institute Gazette**, introduced in 1866, was a multilingual newspaper and the mouthpiece of the Scientific Society, gaining wide popularity for its influential editorials by Sir Syed and shaping Urdu journalism.
    - **Tahzib-ul-Akhlaq**, launched in December 1870, was a monthly journal dedicated to Sir Syed's educational, ethical, and social reform agenda, and played a crucial role in disseminating his progressive ideas for the upliftment of Indian Muslims.

So, only three of the above periodicals were written by Sir Syed Ahmad Khan.
Therefore, option (b) is the correct answer.

Knowledge Box
• **Raja Ram Mohan Roy (1772–1833)** was a prominent Indian social reformer, scholar, and founder of the Brahmo Samaj, a socio-religious reform movement. • His contributions spanned various aspects of Indian society, including religion, social justice, education, and journalism. • **Mirat-ul-Akhbar and Sambad Kaumudi** were both written by Raja Rammohan Roy.

Q23.
Answer: b
Explanation:

- India's Initial Adaptation Communication was submitted to United Nations Framework Convention on Climate Change (UNFCCC) in December 2023.
    - The Adaptation Communication provides a detailed description of India's vulnerability to climate change, its adaptation requirements, and actions being taken or envisaged.
- **The Adaptation Communication was established by Article 7 of the Paris Agreement**. Each Party should submit and update periodically an adaptation communication, which may include information on its priorities, implementation and support needs, plans and actions. **So, statement 1 is correct.**
    - The preparation and submission of an Adaptation Communication are completely voluntary.
- The adaptation communication shall be submitted as a component of or in conjunction with other communications including a national adaptation plan, a national communication, a nationally determined contribution, or a biennial transparency report.
- India's Initial Adaptation Communication to the United Nations Framework Convention on Climate Change (UNFCCC) reveals that **adaptation-related expenditure has grown from 3.7% of GDP in 2015-16 to 5.6% in 2021-22.** It indicates that the total adaptation relevant expenditure for the year 2021-22 was 5.6 per cent of the Gross Domestic Product (GDP), growing from a share of 3.7 per cent in 2015-16. This was up from about **Rs 5 lakh crore in 2015-16 to Rs 13.35 lakh crore in 2021-22.** It shows that the Government has been making consistent efforts to integrate climate resilience and adaptation into development plans and spending a significant amount of resources for adaptation, despite the competing demands especially from the social sector for resources. **So, statement 2 is correct.**
- **Nationally Determined Contributions, or NDCs, are countries' self-defined national climate pledges** under the Paris Agreement. It deals with what they will do to help meet the global goal to pursue 1.5°C. India submitted its first Nationally Determined Contribution (NDC) in the year 2015. In August 2022, India updated its NDC.
    - **Under updated NDC, India pledges to reduce emissions intensity of its GDP by 45 percent by 2030 from 2005 level**, and to achieve 50 percent cumulative electric power installed capacity from non-fossil fuel-based energy resources by 2030. **So, statement 3 is not correct**.

Therefore, option (b) is the correct answer.
Relevance: India is in the final stages of preparing its first ever Biannual Transparency Report (BTR)

Q24.
Answer: c
Explanation:

- Recently, OpenAI released two new Artificial Intelligence (AI) reasoning models named OpenAI o3 and o4-mini.
- According to Open AI, these are reasoning models, that "thinks before they answers" and "can produce a long internal chain of thought before responding to the user". These models are trained to think for longer before responding.
- **In training both models, the company used "reinforcement learning",** a technique previously used by other AI companies, including the Chinese startup DeepSeek. **So, statement 1 is correct.**
    - Reinforcement learning (RL) is a machine learning (ML) technique that trains software to make decisions to achieve the most optimal results. It mimics the trial-and-error learning process that humans use to achieve their goals. Software actions that work towards your goal are reinforced, while actions that detract from the goal are ignored.
- Artificial Intelligence (AI) models o3 and o4-mini were **developed by OpenAI. So, statement 2 is correct.**
- o3 and o4-mini are reasoning models. Reasoning models arrive at answers to user queries in complex and multiple ways. Through "reasoning", the models consider different approaches and solutions to a prompt, while recognising patterns to arrive at the answer.

Therefore, option (c) is the correct answer.
Relevance: OpenAI's 'smartest' new o3 and o4-mini models have been launched

Q25.
Answer: a
Explanation:

- Six "rainbow" herbicides, commonly called Agents Orange, Green, Pink, Purple, White, and Blue, were sprayed on wetlands, rice paddies, forests, mangroves, bamboo groves etc by USA during the Vietnam War.
- **Agent Orange was a toxic herbicide** that the United States used during the Vietnam war. The chemical comprised a 50-50 **mixture of two herbicides — 2,4-D and 2,4,5-T — and a toxic contaminant, known as dioxin. Agent Orange refers to a specific blend of herbicides** used during the Vietnam War. The name "Agent Orange" came from the identifying orange stripe around the 55-gallon drums in which the herbicide was stored.
    - Agent Orange was a blend of herbicides that US forces sprayed over Vietnam to defoliate trees and shrubs, and kill food crops that provided cover and food to North Vietnamese and Viet Cong troops.
    - The severe impact of Agent Orange has been suffered by not only those who were exposed to it during the war, but also the future generations.
    - **Dioxins are highly toxic** and can cause reproductive and developmental problems, damage the immune system, disrupt the endocrine system and cause cancer.
    - Agent Orange was used against broadleaf plants and woody shrubs and trees, including mangroves. **So, pair 1 is correctly matched.**

- **Agent Blue is one of the mixtures of herbicides** that was used by the United States during the Vietnam War. **Agent Blue is an arsenic-based herbicide**. Agent Blue was a mixture of the organoarsenic compounds cacodylic acid (pictured) and sodium cacodylate. When it biodegrades arsenic is released.
    - Agent Blue was sprayed on mangrove forests and rice paddies to destroy them. Agent Blue was effective against grasses and grains, such as rice.
    - **Agent Blue has no half-life** – in other words, it lasts forever in soil, sediments, rivers, canals and public water supplies. Once it is in the environment, its toxicity is magnified as it moves up the food chain, slowly killing and disabling humans as it accumulates in the body. **So, pair 2 is not correctly matched.**
- **Agent Green was a herbicide** and defoliant used by the U.S. military during the Vietnam War. It was one of the so-called "rainbow herbicides." Agent Green's **only active ingredient was 2,4,5-trichlorophenoxyacetic acid (2,4,5-T)**, one of the common phenoxy herbicides of the era. **It contained many times the average level of dioxin found in Agent Orange. So, pair 3 is not correctly matched.**

Therefore, option (a) is the correct answer.
Relevance: April 30th 2025 marks the 50th anniversary of the end of the Vietnam War.

Q26.
Answer: c
Explanation:

- The Global Wind Report is an annual report of **the Global Wind Energy Council (GWEC). So, statement 1 is not correct.**
- The 2025 report was recently published on April 23, 2025.
- According to 2025 report, **the year 2024 was the record-breaking year for the sector which saw 117 gigawatts of new wind capacity installed worldwide** - a marginal increase from the 116.6 GW added in 2023. Total global wind power capacity now stands at an impressive 1,136 GW, demonstrating significant progress in the energy transition. **So, statement 2 is correct.**
- According to report, regional disparities in development continue to shape the global landscape, with **China maintaining its dominant position by accounting for a staggering 70 per cent of all new installations in 2024. So, statement 3 is correct.**
- The United States, Brazil, India and Germany rounded out the top five markets.
- According to report, **the offshore wind sector experienced a concerning setback, with only 8 GW of new capacity installed globally in 2024** — representing a 26 per cent decline from the previous year and the lowest figure since 2021. **So, statement 4 is not correct.**
- The 2025 reveals that current growth projections show the industry will deliver only 77 per cent of the installed capacity required by 2030 to maintain a viable pathway to net-zero emissions. This shortfall threatens international efforts to limit global temperature rise to 1.5°C above pre-industrial levels.

Therefore, option (c) is the correct answer.
Relevance: Global Wind Report published on April 23, 2025

Q27.
Answer: c
Explanation:

- **Sarhul** is a **tribal spring festival** celebrated in **Jharkhand** and the **Chhotanagpur plateau**, symbolising the **union of the Sun and Earth** and marking the **worship of the Sal tree**, believed to be the abode of **Sarna Maa**, the village deity. **So, statement 1 is correct.**
- **Sarhul** literally means **"worship of the Sal tree"** and signifies the **union of the Sun and the Earth**, which is central to tribal cosmology.
- The festival is celebrated by **Adivasi communities** in **Jharkhand**, especially the **Munda, Oraon, Ho, Santal** and **Khadia** tribes and is deeply connected with **sacred groves** locally called **Sarna**, where rituals and offerings are made. **So, statement 2 is correct.**

Therefore, option (c) is the correct answer.

Knowledge Box
Key Facts about Sarhul Festival: • The festival lasts **three days**, involving **rituals at Sarna Sthals (Sacred Groves)**, **offerings of Sal flowers**, **community feasting** and **traditional performances**. • It has also spread to **Assam, Andaman & Nicobar Islands, Nepal, Bangladesh and Bhutan** due to tribal migration. • Sarhul has become a symbol of **Adivasi identity and resistance**, with demands for official recognition of the **Sarna religion** in India's census. • The festival highlights **India's tradition of sacred groves**, which serve as **ecological hotspots** and are known by various regional names like **Devrai (Maharashtra)**, **Devarakadu (Karnataka)**, **Law Kyntang (Meghalaya)**, etc.

Relevance: Sarhul is a traditional festival celebrated by Adivasi communities in Jharkhand and the Chhotanagpur region, marking the onset of spring and the beginning of the new year.

Q28.
Answer: c
Explanation:

- The **2025 Technology and Innovation Report,** released by the **United Nations Conference on Trade and Development (UNCTAD),** highlights that India ranked 36th in 2024 on the Readiness for Frontier Technologies index—an improvement from 48th in 2022. The index evaluates countries based on key pillars including ICT deployment, skills, R&D activity, industrial capacity and access to finance.
- India's performance varied across these dimensions, ranking **3rd in R&D, 10th in industrial capacity, 70th in access to finance, 99th in ICT deployment** and **113th in skills.** The report also noted improvements in human capital indicators—such as increased years of schooling and higher proportions of high-skill employment—in countries like India, Bhutan, Morocco, the Republic of Moldova, and Timor-Leste.

Therefore, option (c) is the correct answer.
Relevance: India has been ranked 10th in the world with significant private investments in Artificial Intelligence (AI) in 2023, according to the 2025 Technology and Innovation Report.

Q29.
Answer: d
Explanation:

- A **Geographical Indication (GI) tag** identifies a product as originating from a specific location and possessing qualities or a reputation attributable to that origin. It acts as a form of intellectual property protection, ensuring that only those producers from the designated region can use the product name, preventing unauthorized use and protecting the authenticity of traditional goods.

Following products have been granted the **Geographical Indication (GI) tag:**

- **Gir Kesar mango** is grown in **Junagadh and Amreli districts** of Gujarat near the Gir sanctuary. It is known for its **distinct taste, aroma and superior quality**. The GI tag ensures only mangoes from this region can be called **Gir Kesar**, preventing misuse and providing **legal protection** under the Geographical Indications Act, 1999. **So, point 1 is correct.**
- **Amalsad chikoo (sapodilla)** from **South Gujarat** is noted for its **exceptional sweetness, fine texture** and **long shelf life**. It is cultivated in **Gandevi, Jalalpore, and Navsari talukas**, contributing around **30% of Gujarat's chikoo production**. The GI tag highlights its **traditional farming practices** and enhances the reputation of **local horticulture**. **So, point 2 is correct.**
- **Kutchi Kharek** is the **indigenous date variety** of **Kutch**, Gujarat, has received the GI tag for its **unique quality and adaptation** to the **semi-arid climate**. It is the **second fruit from Gujarat** after Gir Kesar to receive such recognition, boosting the **livelihoods of local farmers**. **So, point 3 is correct.**
- **Coorg cardamom** is also known as **Coorg green cardamom**, is grown in **Kodagu (Coorg)**, Karnataka. It has a **distinct aroma, faint green color**, and **small size** due to the region's **Western Ghats climate**. It helps protect its **regional identity** and supports the **spice industry's exports**. **So, point 4 is correct.**

So, all four of the above products have been granted the Geographical Indication (GI) tag. Therefore, option (d) is the correct answer.

Knowledge Box
Key Facts about GI Tags in India: ● GI Tags are governed by the **Geographical Indications of Goods (Registration and Protection) Act, 1999**. ● They confer **legal protection**, prevent **unauthorized use** and promote **rural development and traditional knowledge**. ● India's **first GI-tagged product** was **Darjeeling Tea**. ● The GI Registry is based in **Chennai** and is overseen by the **Department for Promotion of Industry and Internal Trade (DPIIT)**. ● GI Tags can apply to **agricultural products, foodstuff, handicrafts, industrial products, and natural goods**.

Q30.
Answer: d
Explanation:

- **Gas hydrates** are crystalline solids composed of gas molecules, usually methane, trapped within a lattice of water molecules. They are considered a **potential source of unconventional energy** and are typically found in **high-pressure, low-temperature environments**, such as the **continental margins under the sea floor**.

- Gas hydrate reserves have been discovered along the **eastern coast**, especially in the **Krishna-Godavari Basin**, but gas hydrates are **not exclusive** to this region. Extensive surveys by India's National Gas Hydrate Program (NGHP) have confirmed the presence of gas hydrates along **both the eastern and western continental margins**, including the **Kerala-Konkan Basin** and **Andaman offshore areas**. **So, statement 1 is not correct.**
- Gas hydrates are generally found in the **shallow sediments of continental slopes and rises**, below the seafloor, in regions where **low temperatures and high pressures** are maintained. These environmental conditions are usually met in **continental margins**, which are transitional zones between continental and oceanic crust. In India, such conditions prevail along both coasts. **So, statement 2 is correct.**

So, statement–I is incorrect, but Statement–II is correct.
Therefore, option (d) is the correct answer.
Relevance: India Energy Week press conference discusses India's gas hydrates potential, challenges, and slow progress in extraction.

Q31.
Answer: b
Explanation:

- The Government of India has signed various water treaties/ agreements with the neighbouring countries for mutual benefits in the field of water resources.
- **The Indus Waters Treaty (IWT) was signed by India and Pakistan** on September 19, 1960, to determine the distribution of the waters of the Indus and its tributaries. It was brokered by the World Bank and signed in Karachi by then Indian Prime Minister Jawaharlal Nehru and then Pakistan President Ayub Khan. The IWT established a framework which enabled the equitable administration of the Indus River system regardless of political tensions between India and Pakistan.
    - The terms of the treaty gave India full control of the **eastern basin rivers including Ravi, Beas, and Sutlej** whereas Pakistan received exclusive access to western basin rivers namely Indus, Jhelum, and Chenab. **All the waters of the Eastern Rivers were allocated to India for her unrestricted use. So, pair 1 is correctly matched.**
    - The "Permanent Indus Commission" is an essential element of the treaty through which both sides exchange information and data, and settle minor disputes. It forms part of an established three-step dispute resolution framework. The two other dispute resolution mechanisms are the World Bank-appointed Neutral Expert and, as a last resort, the Permanent Court of Arbitration.
- **The Ganga Water Treaty** was signed on December 12, 1996 by then Indian Prime Minister H D Deve Gowda and his Bangladeshi counterpart Sheikh Hasina. The treaty ensured the fair sharing of river water after the construction of the Farakka Barrage.
    - The 1996 Ganga Water Sharing Treaty is an agreement between India and Bangladesh to share the waters of the Ganga River at the Farakka Barrage in West Bengal for 30 years. The treaty expires in 2026 and can be renewed by mutual consent.
    - It **provides for sharing of Ganga waters at Farakka**. As per the Treaty, the Ganga waters is being shared at Farakka (which is the last control structure on river Ganga in India) during lean period, from 1st January to 31st May every year, on 10-day period basis as per the formula provided in the Treaty. **So, pair 2 is correctly matched.**
- A Treaty has been signed with Nepal in the year 1996, viz. Mahakali Treaty. Under the Mahakali Treaty of 1996, multiple agreements were combined for Integrated Development of the Mahakali River (known as river 'Sarda' in India), including Sarda Barrage, Tanakpur Barrage and Pancheshwar Project.

- **Indo-Bhutan** hydropower cooperation began in 1961 with the signing of the **Jaldhaka Agreement. The Jaldhaka Project is situated on the Indian side of Indo-Bhutan border** in West Bengal. The major part of power produced at Jaldhaka hydropower plant was exported to southern Bhutan. **So, pair 3 is not correctly matched.**

Therefore, option (b) is the correct answer.

Relevance: The location of China's proposed dam on the Yarlung Tsangpo River poses a risk of disrupting the river's natural flow.

Q32.

Answer: c

Explanation:

- Carbon Credit Trading Scheme aims to create an emissions trading system by providing a market for trading in greenhouse gas (GHG) emissions. It will develop an Indian Carbon Market (ICM) for trading of the Carbon Credit Certificates.
- The Central Government has notified the Carbon Credit Trading Scheme in exercise of the powers conferred by clause (w) of section 14 of the Energy Conservation Act, 2001.
- The CCTS was launched **to create a framework for the trading of carbon credits,** to facilitate the reduction of emissions in energy intensive industries.
- The CCTS consists of a compliance mechanism, **mandating emissions intensity targets for high-emission industry sectors.** It also has an offset mechanism, allowing voluntary emissions reduction projects.
- Under the compliance mechanism, the Central Government shall specify the registered entities as obligated entities. The obligated entities shall comply with the prescribed GHG emission intensity targets in each compliance year. The obligated entities who reduce their GHG emission intensity below the target GHG emission intensity shall be eligible for issuance of Carbon Credit Certificates and the entities who are not able to achieve the target will be required to surrender/purchase equivalent number of certificates based on shortfall.
- India's **Carbon Credit Trading Scheme (CCTS) is based on intensity of emission** that suits a developing nations model as against the system in Western countries that deals with **absolute emission**. **So, statement 1 is correct.**
- **Under the offset mechanism, the non-obligated entities** can register their projects for GHG emission reduction or removal or avoidance for issuance of Carbon Credit Certificates. In voluntary trading, **there is no externally imposed limit on carbon emissions.** Rather, a company may choose to purchase carbon credits to lower its net emissions. **So, statement 2 is correct.**

Therefore, option (c) is the correct answer.

Relevance: The Draft Greenhouse Gases Emissions Intensity (GEI) Target Rules, 2025, notified by the Ministry of Environment, Forest and Climate Change on April 16, puts in place a compliance mechanism for the Carbon Credit Trading Scheme, 2023 (CCTS).

Q33.

Answer: a

Explanation:

- The National Centre for Polar and Ocean Research (NCPOR) is the **nodal agency for India's research in the Polar and Southern Ocean regions**, conducting expeditions to the **Arctic, Antarctica, Himalayas, and Southern Ocean**, and overseeing related logistics and scientific research. **So, statement 1 is correct.**
- NCPOR functions under the **Ministry of Earth Sciences (MoES)** and not the Ministry of Environment, Forest and Climate Change. **So, statement 2 is not correct.**

- It plays a critical role in India's **Deep Ocean Mission**, focusing on **deep-sea mining, submersibles, and exploration of rare minerals and metals** in the Indian Ocean.
- Established in **1998**, NCPOR supports projects like **EEZ mapping and continental shelf studies**, which have strategic importance.
- It was formerly known as **National Center for Antarctic and Ocean Research (NCAOR)** and is located in **Vasco da Gama, Goa**.

Therefore, option (a) is the correct answer.

Relevance: The National Centre for Polar and Ocean Research (NCPOR) is building a new ocean research vessel to explore the deep Indian Ocean for rare minerals and metals as part of India's Deep Ocean Mission.

Q34.

Answer: c

Explanation:

- The **Farm Machinery Banks (FMB) scheme** aims to make modern agricultural machinery accessible to small and marginal farmers by promoting group ownership and custom hiring services. Implemented under the larger **Sub-Mission on Agricultural Mechanization (SMAM**), it seeks to bridge the mechanization gap in rural India by offering financial support to Farmer **Producer Organizations (FPOs), Self Help Groups (SHGs), and Cooperative Societies. So, statement 1 is correct.**
- The scheme is not limited to individual farmers. In fact, the core focus is on **group-based beneficiaries such as FPOs, SHGs, and PACs.** It encourages collective ownership and shared usage, especially in regions with low mechanization levels. **So, statement 2 is not correct.**
- A key feature of FMBs is the promotion of **Custom Hiring Centers (CHCs) at the village level.** These CHCs enable smallholders to rent expensive machinery like combine harvesters, rotavators, and power tillers without buying them. **So, statement 3 is correct.**

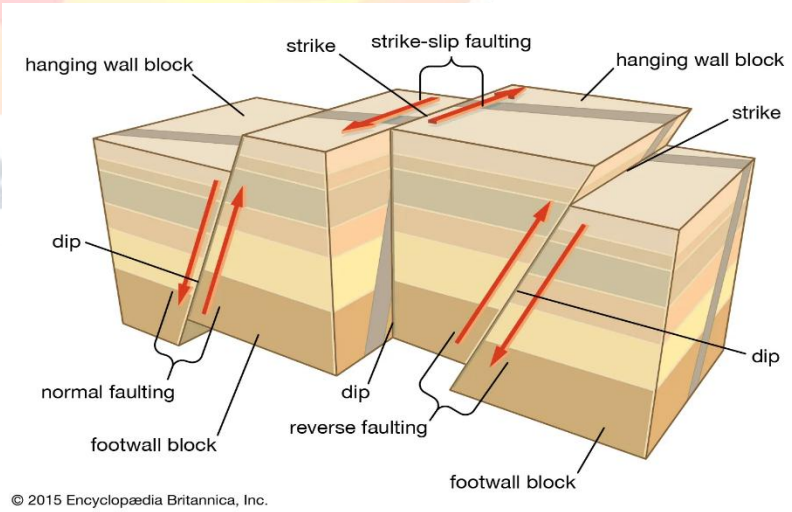
Therefore, option (c) is the correct answer.

Q35.

Answer: a

Explanation:

- **Strike-slip faults** are typically **transform boundaries**, where horizontal motion dominates (e.g., **San Andreas Fault**, **Anatolian Fault**).
- A **strike-slip fault** occurs when two blocks of the Earth's crust **slide past each other horizontally**, with little or no vertical movement. A well-known example is the **San Andreas Fault** in California. **So, statement 1 is correct and statement 2 is not correct.**



- These faults are known to produce **strong earthquakes** due to the **frictional buildup** and sudden release of energy along the fault line.
- The **2023 Turkey-Syria earthquakes**, including a **magnitude 7.8** and a **7.5** aftershock, were caused by a **strike-slip fault** between the **Arabian Plate** and the **Anatolian Plate**.

Therefore, option (a) is the correct answer.

Q36.
Answer: a
Explanation:

- **Project Seabird** refers to the establishment of **INS Kadamba**, a large and modern naval base developed by the **Indian Navy at Karwar**, on the **western coast of India in Karnataka**. It is not associated with nuclear submarine operations. **So, pair 1 is not correctly matched**.
- **Project Varsha** is related to the development of a **strategic naval base near Visakhapatnam on the eastern coast**, primarily to **support India's fleet of nuclear-powered submarines**. The facility is intended to strengthen India's **second-strike nuclear capability** under its nuclear doctrine. The base at Karwar is instead part of **Project Seabird**, not Varsha. **So, pair 2 is not correctly matched.**
- **Project VIJAYAK** is a key **Border Roads Organisation (BRO)** initiative, undertaken to develop and maintain road infrastructure in the **Kargil sector and surrounding regions**, including **Zanskar** in **Ladakh**. The project gained strategic significance particularly after the **Kargil War of 1999**, highlighting the need for improved military and civilian connectivity in this sensitive border region. **So, pair 3 is correctly matched.**

So, only one of the pairs given above are correctly matched.
Therefore, option (a) is the correct answer.
Relevance: The Defence Minister of India will inaugurate two big piers and seven towers with 320 homes for Navy officers and Defence civilians as part of Project Seabird at Naval Base Karwar in Karnataka.

Q37.
Answer: a
Explanation:

- **Regional Rural Banks (RRBs)** were established in India in 1975 under the provisions of the **Regional Rural Banks Act, 1976**, based on the recommendations of the **Narasimham Committee**. Their primary objective is to provide **credit and other financial services to the rural population**, especially to **small and marginal farmers, artisans and agricultural laborers**. The ownership of an RRB is divided among:
    - **Central Government – 50%**
    - **Sponsoring Public Sector Bank – 35%**
    - **Concerned State Government – 15%**
- The sponsor bank of RRBs can be Public Sector Bank and also Private banks. **Jammu & Kashmir Bank**, a private sector bank, sponsors the **Ellaquai Dehati Bank**. This indicates that private sector banks can also act as sponsor banks for RRBs. **So, statement 1 is not correct.**
- The **Reserve Bank of India (RBI)** and the **National Bank for Agriculture and Rural Development (NABARD)** jointly oversee the functioning of RRBs. While RBI provides the regulatory framework under the Banking Regulation Act, NABARD plays a pivotal role in supervising, monitoring, and providing financial assistance to RRBs. This dual oversight ensures that RRBs operate efficiently and fulfill their mandate of serving rural areas. **So, statement 2 is correct.**
- Although RRBs are primarily established to serve rural and semi-urban populations, they are **not restricted** to these areas. RRBs can operate in urban areas to mobilize deposits and provide banking services. At present RRBs have around **22,000 branches**, with approximately **92% located in rural and semi-urban areas**, implying that about **8%** are in urban regions. **So, statement 3 is not correct.**

Therefore, option (a) is the correct answer.

Q38.
Answer: c
Explanation:

- The **India Skills Accelerator (ISA)** is a forward-looking initiative launched by the **Ministry of Skill Development and Entrepreneurship (MSDE)** in collaboration with the **World Economic Forum (WEF)**.
- This program aims to transform India's skilling ecosystem by fostering **public-private collaboration**, addressing **critical skill gaps** and preparing the Indian workforce for the **future of work**. It responds to the rapidly evolving demands of the global job market, emphasizing **inclusive upskilling**, **reskilling**, **policy reform**, and **institutional innovation**.
- The initiative promotes **cross-sectoral cooperation** and seeks to address the **national skill gap** across all regions and demographics.

Therefore, option (c) is the correct answer.
Relevance: Ministry of Skill Development and Entrepreneurship has collaborated with the World Economic Forum to launch the "India Skills Accelerator" initiative.

Q39.
Answer: d
Explanation:

- The **Genome India Project** is a pioneering initiative launched by the **Department of Biotechnology (DBT), Ministry of Science and Technology, Government of India**, with the aim to **map the genetic diversity** of the Indian population. It seeks to sequence the genomes of at least **10,000 Indian individuals** across various ethnic and regional groups. This initiative is inspired by the success of the **Human Genome Project** and is expected to enhance India's capabilities in **genomic medicine**, particularly in **personalized treatment**, **rare disease diagnosis** and **predictive healthcare**.
- The core objective of the Genome India Project is to develop a **representative reference genome** that captures the **genetic diversity of the Indian subcontinent**, which is one of the most **genetically heterogeneous regions** in the world. **So, statement 1 is correct.**
- **Genetic variants** refer to differences in DNA sequences among individuals. These variations can occur at a **single DNA base pair** or over large segments of chromosomes. Some of these changes are harmless, while others can influence physical traits, metabolism, **drug response** and **risk of developing diseases** such as cancer, diabetes, or cardiovascular disorders. Identifying these variants helps scientists understand **how genetic makeup affects health**, enabling the development of **targeted therapies** and better disease prediction models. **So, statement 2 is correct.**
- **Single Nucleotide Polymorphisms (SNPs)** are the most prevalent form of genetic variation found in the human genome. An SNP is a **variation in a single nucleotide (A, T, C, or G)** that occurs at a specific position in the genome. These variations are often used as **biomarkers** in genomic studies because they can be linked to specific traits or diseases. **So, statement 3 is correct.**

Therefore, option (d) is the correct answer.

Q40.
Answer: d
Explanation:

- Recognizing the importance of inclusive financial education, the **Investor Education and Protection Fund Authority (IEPFA)** under the **Ministry of Corporate Affairs**, in collaboration with the **India Post Payments Bank (IPPB)**, launched the **"Niveshak Didi" initiative**. This initiative seeks to transform rural women into community-level financial educators.

- The program **trains women postal workers and local community leaders** to become **"financial didis"**, who then educate others in their villages on vital financial topics such as budgeting, responsible investing, digital banking and fraud prevention. The initiative leverages **India Post's extensive last-mile connectivity** and the **community trust** women leaders command, especially in rural areas. The long-term goal is to **create a ripple effect of financial empowerment** across underserved regions.

Therefore, option (d) is the correct answer.

Relevance: IEPFA and India Post Payments Bank (IPPB) have signed a Memorandum of Agreement to launch Phase 2 of the *Niveshak Didi* initiative

Q41.

Answer: c

Explanation:

- India's commitment to decentralization and grassroots development is being systematically tracked through innovative tools like the **Panchayat Advancement Index (PAI)** and the **Devolution Index**. These indices have been introduced by the **Ministry of Panchayati Raj** to promote evidence-based planning, assess Panchayat performance and support policy alignment with the goals of **Viksit Bharat** and the **Sustainable Development Goals (SDGs) 2030**.
- The **PAI** is a multi-domain and multi-sectoral index that evaluates the development performance of **over 2.5 lakh Gram Panchayats** across nine thematic areas aligned with **Localized SDGs (LSDGs)**. These include areas such as poverty alleviation, health, women-friendliness, clean water and good governance. With over **435 indicators** and **566 data points**, the PAI offers a granular look at the development status of local communities and provides a **benchmark for progress**, helping to customize Panchayat Development Plans (PDPs) more effectively**. So, statement 1 is correct.**
- The **Devolution Index** is a diagnostic tool that assesses the extent of **functional autonomy** granted to Panchayats by state governments. It reflects the real implementation of **Article 243G** of the Constitution, which empowers states to devolve 29 subjects (as listed in the Eleventh Schedule) to local bodies. The Index considers six critical dimensions—**Framework, Functions, Finances, Functionaries, Capacity Building, and Accountability**—and evaluates how empowered Panchayats are to plan, fund and execute local initiatives. It also serves as a practical roadmap for improving decentralization, strengthening cooperative federalism and promoting accountable governance. **So, statement 2 is correct.**
- One of the key purposes of the **Panchayat Advancement Index** is to enable **evidence-based planning** by helping Panchayats identify their performance across LSDG themes. The index categorizes Panchayats into performance bands—**Achievers, Front Runners, Performers, Aspirants, and Beginners**—allowing them to assess their current status and prioritize future interventions. It also promotes **healthy competition**, encourages capacity building, and provides a **data-driven foundation** for more efficient resource allocation and targeted policy support, particularly in Panchayats lagging in development outcomes.

Therefore, option (c) is the correct answer.

Relevance: The Ministry of Panchayati Raj released the comprehensive Devolution Index Report.

Q42.
Answer: a
Explanation:

- The **IUCN Green Status of Species** is a relatively recent but important addition to global conservation frameworks. While the **IUCN Red List** assesses species based on their risk of extinction, the **Green Status** complements it by focusing on **how much of a species' ecological role has been restored** and how successful conservation actions have been.
- The primary purpose of the **Green Status of Species** is to shift the conservation narrative from solely preventing extinction to also **measuring recovery**. Unlike the Red List, which tells us how close a species is to being extinct, the Green Status tells us **how much of its ecological function and distribution has been restored**. It introduces the concept of a **"Green Score"**, ranging from 0 to 100, which quantifies a species' progress toward full recovery. **So, statement 1 is correct.**
- According to IUCN's definition, a species achieves **"full recovery"** when:
  - It is **present** in all areas of its indigenous range, including parts it no longer currently occupies;
  - It is **viable** (i.e., not at risk of extinction) across that range;
  - It is **performing its ecological functions** (e.g., pollination, predation, seed dispersal) in all parts of that range.
- This is a **rigorous and idealized standard** of recovery. It is not always achievable—especially in regions where ecosystems have been irreversibly altered—but it serves as a consistent **benchmark** to measure progress, guide conservation priorities, and recognize success stories. **So, statement 2 is correct.**
- The **Green Score** is a measure of how much of a species' full ecological recovery has been achieved, **not just its conservation dependence**. While one of the **four conservation metrics—Conservation Dependence**—does address how a species would fare without ongoing conservation, the Green Score itself is a **broader indicator** of recovery across space, viability and function. Moreover, **the Green Score is not limited to Critically Endangered species**. It can be applied to **any species**, regardless of its Red List category. **So, statement 3 is not correct.**

Therefore, option (a) is the correct answer.

Q43.
Answer: c
Explanation:

- The treatment of prisoners is a significant human rights concern globally. To ensure that even those deprived of liberty are treated with dignity, the **United Nations** has developed various international guidelines. Among the most notable are the **United Nations Rules for the Treatment of Women Prisoners and Non-custodial Measures for Women Offenders** (commonly known as the **Bangkok Rules**) and the **United Nations Standard Minimum Rules for the Treatment of Prisoners**, revised and renamed in 2015 as the **Nelson Mandela Rules**. While these frameworks are not legally binding, they serve as moral and operational benchmarks for prison reforms and human rights practices, including in countries like India.

- The **Bangkok Rules**, adopted by the UN General Assembly in 2010, are the first set of international guidelines that specifically address the **needs and rights of women prisoners** and **women offenders**. Recognizing that women face unique vulnerabilities and are often overlooked in prison management systems that are primarily designed for male inmates, the rules provide guidance on **gender-sensitive prison conditions**, **healthcare**, **mental well-being**, **pregnancy and childcare** and **protection from abuse**. They also encourage the use of **non-custodial measures**, such as community-based sanctions, particularly for pregnant women and mothers of young children.
- The **Nelson Mandela Rules**, adopted unanimously by the UN General Assembly in 2015, are a **revised version of the 1955 Standard Minimum Rules for the Treatment of Prisoners**. These rules serve as the **global standard** for humane treatment of prisoners, irrespective of their gender, nationality, or the crime committed. Key principles include **protection from torture**, **access to healthcare**, **education**, **contact with families**, and **safeguards for solitary confinement**. While not binding under international law, they have gained broad acceptance.

Therefore, option (c) is the correct answer.

Q44.
Answer: b
Explanation:

- **Mesoscale Convective Systems (MCSs)** are large **organized groups of thunderstorms** that play a crucial role in weather patterns, especially in **tropical and mid-latitude regions.** MCSs are known for their potential to produce **intense rainfall, lightning, strong winds and occasionally hail**, making them significant from both meteorological and disaster management perspectives.
- MCSs are defined by their **size, structure and longevity**. Typically, they extend over **100 km or more** and last for **at least 6 hours**, although some can persist for longer periods. They often form in environments with strong atmospheric instability and moisture convergence and their structure allows multiple thunderstorms to sustain and organize into a coherent system. The large size and long duration make them capable of producing **significant weather impacts**, such as heavy rain and damaging winds, over broad areas. **So, statement 1 is correct.**
- MCSs are **rare in polar regions** due to the lack of the **heat and moisture** necessary for sustained convection. Instead, they are **most common in tropical, subtropical and mid-latitude areas**, where warm, moist air masses and atmospheric instability are present. In India, MCSs frequently form during the **pre-monsoon and monsoon seasons**, contributing to rainfall events across central, eastern, and northeastern India. **So, statement 2 is not correct.**
- Due to the **intense and prolonged rainfall** associated with MCSs, they are a leading cause of **flash flooding**, especially in **urban and low-lying areas**. The rainfall rates in these systems can exceed **several centimeters per hour**, overwhelming drainage systems and leading to rapid accumulation of water. Flash floods triggered by MCSs are particularly dangerous because they can occur with little warning and have a high potential for **loss of life and property damage**. **So, statement 3 is correct.**

Therefore, option (b) is the correct answer.

Q45.
Answer: b
Explanation:

- The **Reserve Bank of India (RBI)** adjusts its **monetary policy stance**—whether accommodative, neutral, or tightening—based on evolving macroeconomic conditions such as inflation, growth, credit demand and external vulnerabilities. A shift from an **accommodative stance to a neutral stance** signals that the RBI is no longer inclined toward further rate cuts, but also not yet ready to increase interest rates. This change typically occurs when **inflationary pressures begin to surface**, making it necessary for the RBI to maintain flexibility in either direction.
- When inflation begins to rise and approaches or crosses the **upper band of the RBI's inflation target (4% ± 2%)**, the central bank may choose to pause rate cuts to **balance inflation control with the need to support growth**. A neutral stance allows **greater flexibility**, enabling the RBI to respond to inflation risks without abandoning its growth concerns.
- If inflation is low and growth is sluggish, the RBI is more likely to **continue with an accommodative stance** to support economic activity. In this situation, maintaining or even reducing interest rates would be justified to stimulate demand and investment.
- High unemployment and weak credit demand point to **deteriorating growth conditions**, which would usually justify a **continued accommodative stance** or even further rate cuts.
- Falling oil prices and currency appreciation are **disinflationary** forces. If they result in deflationary concerns, the RBI would likely retain an **accommodative stance** or even cut rates further to combat deflation and support domestic demand.

Therefore, option (b) is the correct answer.

Q46.
Answer: b
Explanation:

- The **Africa India Key Maritime Engagement (AIKEYME)** is a significant multilateral naval exercise initiated in **April 2025** as part of India's growing strategic cooperation with **African nations** in the **Indian Ocean Region (IOR)**.
- The **Africa India Key Maritime Engagement (AIKEYME)** is **co-hosted by India and Tanzania**. The inaugural edition of the exercise took place in **Dar-es-Salaam, Tanzania** in **April 2025**. While **South Africa is one of the participating countries**, it is **not a co-host**. **So, statement 1 is not correct.**
- AIKEYME 2025 sees participation from **11 Indian Ocean nations**: **India, Tanzania, Comoros, Djibouti, Eritrea, Kenya, Madagascar, Mauritius, Mozambique, Seychelles** and **South Africa**. This wide representation reflects the inclusive and collaborative nature of the engagement, aiming to bring together diverse regional stakeholders to address common maritime challenges such as illegal fishing, piracy, and trafficking. **So, statement 2 is correct.**
- The exercise aligns with India's **SAGAR** (Security and Growth for All in the Region) vision and **MAHASAGAR** (Mutual and Holistic Advancement for Security and Growth Across Regions) initiative, which the PM unveiled in March 2025. **So, statement 3 is correct.**

- Its core objective is to **foster multilateral cooperation** in tackling **regional maritime security threats** and to **improve interoperability among participating naval forces.** Through structured **harbour phases** (which include training, seminars, and interaction) and **sea phases** (focused on joint operational drills), the exercise promotes trust, coordination, and capability-building in a region increasingly central to global trade and security.

So, only two of the above statements are correct.
Therefore, option (b) is the correct answer.
Relevance: Recently, the inaugural edition of the Africa India Key Maritime Engagement (AIKEYME), a large-scale multilateral maritime exercise held in Tanzania.

Q47.
Answer: b
Explanation:

- **Pulikali** (from Kerala) and **Pulivesham** (from Andhra Pradesh and parts of Tamil Nadu) stand out as iconic **tiger-themed performances** that captivate audiences with their visual splendor, rhythmic energy and symbolic power. Both art forms have deep roots in **festive celebrations** like Onam and Dussehra and reflect themes of **bravery, vigor and community celebration**.
- **Pulikali**, meaning "Tiger Dance," is a well-known folk art from **Kerala**, especially performed on the **fourth day of the Onam festival**. Artists paint their bodies in bright **yellow, red, and black tiger stripes** and dance to traditional drums like the **chenda**, **thakil**, and **udukku**.
- **Pulivesham** is a similar tiger-themed dance found in **Andhra Pradesh and Tamil Nadu**, usually performed during **Dussehra or Muharram**. It is a **one-man act**, where the dancer, painted like a tiger and often supported by another person managing a long artificial tail, mimics tiger-like movements and sometimes lifts **brass pots with his teeth** to symbolize strength. This folk act is accompanied by rhythmic instruments like the **dappu** or **mridangam**, giving it a unique regional flavor. Both traditions are symbolic of power, festivity, and community entertainmen

Therefore, option (b) is the correct answer.

Q48.
Answer: a
Explanation:
Aerial refueling is the process of transferring fuel from one aircraft (the tanker) to another aircraft (the receiver).

- Refuelling can take place both on the ground and in mid-air.
- Ground refuelling is the most common form of aircraft refuelling and involves transferring fuel while the aircraft is stationary. There are two primary methods of ground refuelling:
    - **Gravity Refuelling:** Ideal for smaller aircraft, **this method relies on the force of gravity to transfer fuel from a tanker into the aircraft's fuel tanks.** Although it's a slower process, gravity refuelling is straightforward and doesn't require complex equipment, making it a cost-effective option.
    - **Pressure Refuelling:** Commonly used for larger commercial and military aircraft, pressure refuelling utilises a pump to force fuel into the aircraft's tanks.
- In the **buddy-buddy method, two fighter aircraft fly together, with one serving as the tanker and the other as the receiver. So, pair 1 is correctly matched.**

- Aerial refuelling, or in-flight refuelling, is an advanced process where fuel is transferred from one aircraft to another while in flight. The two primary methods of aerial refuelling are:
    - **Probe drogue system: It is an aerial refuelling technique.** It involves the deployment of tension controlled fuel hose with attached drogue coupling **from tanker aircraft and receiver aircraft receives** it by maintaining the proper altitude. There is a simple fuel pressure assisted spring loaded mechanical latching mechanism in the drogue that holds the probe in place during refueling. On completion of refueling process receiver aircraft simply drops back to detach the probe. **So, pair 2 is not correctly matched.**
    - In the **Flying Boom System,** the receiving plane flies in close formation to the tanker. A boom operator in the tanker then flies a rigid boom into a receptacle on the top of the aircraft. In this technique**, a rigid, telescopic boom is used**, which is manoeuvred by a boom operator on the tanker aircraft to connect to the receiving aircraft. **So, pair 3 is not correctly matched.**
        - Probe drogue system is a flexible and widely adopted technique that allows multiple aircraft to be refuelled simultaneously.

So, only one of the pairs given above is correctly matched.
Therefore, option (a) is the correct answer.
Relevance: India has inked a Rs 63,000 crore deal with France to procure 26 Rafale-Marine (Rafale-M) fighter jets for the Navy

Q49.
Answer: b
Explanation:

- The **SolarX** and **STAR-C** initiatives are flagship programs launched by the **International Solar Alliance (ISA)**, an international treaty-based organization with the mission to promote solar energy deployment globally, especially in **sunshine-rich but resource-constrained developing countries**. These initiatives aim to foster **innovation**, **capacity building** and **institutional support** for solar energy adoption across ISA member countries.
- The **SolarX Startup Challenge** is an innovation-driven initiative launched by the **ISA** in collaboration with Invest India and other partners. Its objective is to **identify and support scalable, investable and impactful solar energy startups** across the globe. The initiative focuses on:
    - Promoting **entrepreneurship** in clean energy.
    - Offering **financial assistance**, **mentorship**, and **market access** to selected startups.
    - Addressing **energy access challenges** in underserved areas through innovative, solar-based business models.
- The **STAR-C initiative** (Solar Technology and Application Resource Centres) is a strategic partnership between the **ISA and the United Nations Industrial Development Organization (UNIDO)**. The program focuses on **building institutional and human capacity** in member countries for widespread adoption of solar energy. Its goals include:
    - Establishing **regional training hubs** and **centres of excellence** for solar energy technologies.
    - Conducting training and certification programs for **technicians, engineers, entrepreneurs, and policymakers**.

- Promoting **standardization, quality assurance, and skill development** across the solar value chain.

Therefore, option (b) is the correct answer.
Relevance: India is considering expanding its Solar Technology and Application Resource Centres (STAR -C) initiative to a number of Pacific Island countries.

Q50.
Answer: d
Explanation:

- **Wet-bulb Temperature** is the lowest temperature to which air can be **cooled by the evaporation of water into the air at a constant pressure.** It is measured by covering a thermometer bulb with a wet cloth and letting the water evaporate. As the water evaporates, it cools the thermometer, showing the wet bulb temperature.
- A wet-bulb temperature of **35°C** is considered the **maximum limit of humidity** that humans can handle. Beyond this, the body can no longer effectively cool itself via perspiration.
- The **wet bulb temperature is always lower than the dry bulb temperature** but will be identical with 100% relative humidity (the air is at the saturation line).
- The dry bulb temperature is the ambient temperature. The difference between wet and dry bulb temperatures is a measure of the humidity of the air.
- **The Dew Point is the temperature at which water vapor starts to condense out of the air,** the temperature at which air becomes completely saturated. Above this temperature the moisture will stay in the air. The dew point is the temperature and pressure at which the first drop of water vapor condenses into a liquid.

Therefore, option (d) is the correct answer.
Relevance: According to a new study published in Nature Communications, rainfall and moisture conditions in the soil may influence the onset of extreme humid heat in the global tropics and subtropics, including India.

Q51.
Answer: b
Explanation:

- **'*Typhloperipatus williamsoni*'** is a unique velvet worm species belonging to the phylum **Onychophora**, which is often regarded as one of the **oldest living fossil groups** on Earth.
- '**Living fossils**' is a term used to describe isolated, anomalous species of organisms which have largely retained their ancestral characteristics and serve as a link between two different groups of organisms, which are derived from a common ancestor. These living fossils exhibit characteristics of the ancient now extinct common ancestor and have amalgamation of features of both the groups. Some examples of living fossils include Lung fishes (living fossil between Fishes and Amphibians), coelacanths (between vertebrates and invertebrates) etc.
- Velvet worms (Onychophora) are ancient invertebrates, believed to be **over 350 million years old**, predating the age of dinosaurs. They are part of a very small and old phylum with only about **200 described species worldwide**, grouped into just two families. These exhibit characteristics of both **Arthropods and Annelids** hence is considered a connecting link and a living fossil between the two.
- *T. williamsoni* was first discovered in 1911 in the Upper Siang Valley, since then there has been no reported sighting of the organism, till the recent one in April 2025.

- Molecular studies reveal that **South Asian Onychophorans diverged from their South American and African relatives around 237 million years ago**, and interestingly, they are not closely related to Australian Onychophorans, which is highly unusual for invertebrates in this region. This suggests a unique evolutionary path, emphasizing its **paleontological importance and biogeographical rarity**.

Therefore, option (b) is the correct answer.

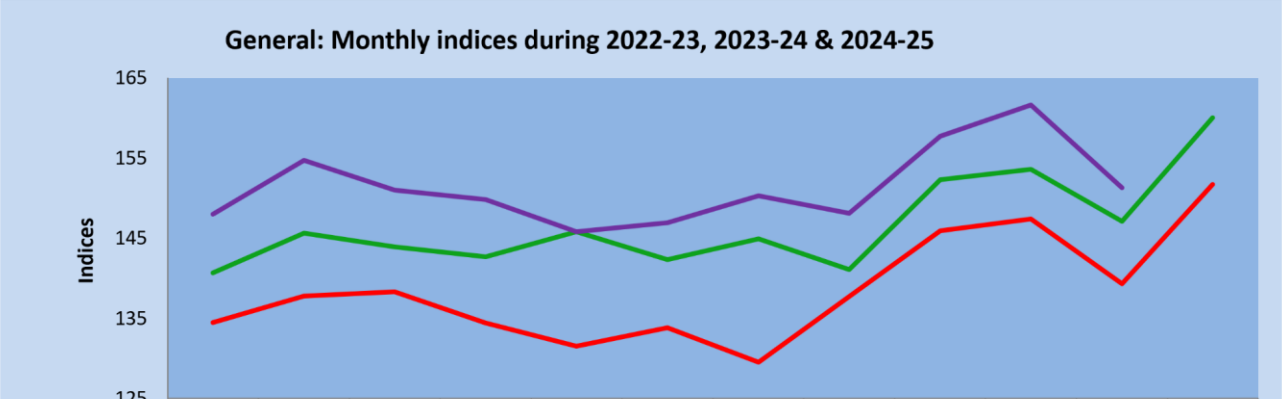
Relevance: A team of researchers recently reported rediscovering a long-lost species of the velvet worm (phylum Onychophora), named Typhloperipatus williamsoni

Q52.

Answer: c

Explanation:

- The **Index of Industrial Production (IIP)** is a crucial indicator that measures the performance of various sectors of the Indian economy, particularly **manufacturing, mining, and electricity**. It is a **composite indicator** that tracks the **volume of production** in different industries within the economy over a specific period.
- The IIP is released by the **Central Statistics Office (CSO)**, which functions under the **Ministry of Statistics and Programme Implementation (MoSPI)**. The CSO compiles and publishes the data on a monthly basis. **So, statement 1 is correct.**
- The current **base year** used for calculating the IIP is **2011–12**, which replaced the earlier base year of 2004–05. The change in base year is part of the periodic revision process to reflect structural changes in the economy and keep the index relevant. The base year revision also aligns the IIP with other key macroeconomic indicators like the GDP and the Wholesale Price Index (WPI), which also use 2011–12 as their base year. **So, statement 2 is correct.**
- According to the official data published for **February 2025**, the **IIP index** stands at **151.3**, which is **higher** than both **February 2024 (147.1)** and **February 2023 (139.3)**.



	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec*	Jan*	Feb*	Mar
2022-23	134.5	137.8	138.3	134.4	131.5	133.8	129.5	137.7	145.9	147.4	139.3	151.7
2023-24	140.7	145.6	143.9	142.7	145.8	142.3	144.9	141.1	152.3	153.6	147.1	160.0
2024-25	148.0	154.7	151.0	149.8	145.8	146.9	150.3	148.1	157.7	161.6	151.3	

*The Figures December 2024, January 2025 and February 2025 provisional

- This clearly indicates a **positive year-on-year growth trend** since the last 2 years in industrial activity despite the fact that the **growth rate** for February 2025 was reported at **2.9%**, the slowest in six months.

Therefore, option (c) is the correct answer.

Relevance: The Index of Industrial Production (IIP) grew at its slowest rate in six months, clocking a pace of 2.9% in February 2025.

Q53.
Answer: c
Explanation:

- In a landmark move for international climate policy, the **International Maritime Organization (IMO)**, a specialized UN agency, adopted the **world's first global carbon tax on the shipping industry** in April 2025.
- This decision, supported by **63 countries including India, China, and Brazil**, is aimed at reducing greenhouse gas emissions from ships and promoting a shift to **cleaner technologies and fuels**.
- The move was part of ongoing efforts to **decarbonise the maritime sector**, which accounts for nearly 3% of global emissions.
- Starting in 2028, ships will either need to switch to **low-emission fuels** or **pay fees** based on the intensity of their emissions. Although hailed as a historic step, the policy has faced criticism for **falling short of emission targets** and **excluding broader climate finance commitments**.

Therefore, option (c) is the correct answer.
Relevance: India and 62 other countries voted in favour of the world's first-ever global carbon tax imposed on the shipping industry

Q54.
Answer: c
Explanation:

- The **Karlsruhe Tritium Neutrino (KATRIN)** experiment is one of the most advanced efforts in experimental particle physics aimed at solving a fundamental question **regarding *the mass of the neutrino*.**
    - Neutrinos are electrically neutral, weakly interacting elementary particles produced abundantly in the universe—during radioactive decay, in nuclear reactions in stars like the sun, and in supernovae. Though they were long believed to be massless, recent studies, including those by KATRIN, have **confirmed that they do possess a tiny, but non-zero, mass.**
- The KATRIN experiment is specifically designed to **measure the mass of the electron antineutrino**—the antimatter counterpart of the electron neutrino. To do this, it examines **beta decay** of **tritium**, a radioactive isotope of hydrogen. In this decay process, a tritium nucleus transforms into a helium-3 nucleus, releasing an electron and an **electron antineutrino**. By precisely analyzing the energy spectrum of the emitted electrons, especially near the high-energy endpoint of the spectrum, KATRIN is able to detect the small effect of the antineutrino's mass. **So, statement 1 is correct.**
- The mass of the neutrino plays a critical role in the **endpoint energy** of the emitted electrons in beta decay. In a scenario where the neutrino is massless, the electron can attain a specific maximum energy. However, if the neutrino has mass, it carries away some of the energy, reducing the maximum energy that the electron can possess. This means that a **nonzero neutrino mass slightly shifts and modifies the shape of the energy spectrum** of the electrons emitted in beta decay. The KATRIN experiment uses this principle to set **upper limits** on the neutrino mass.
- In its latest result (April 2025), KATRIN has reported that the **mass of the electron antineutrino is less than 0.45 electron volts**, improving its previous upper limit by nearly half. This precise

measurement, based on over **36 million decay events**, makes KATRIN one of the most sensitive tools in neutrino physics. **So, statement 2 is correct.**

Therefore, option (c) is the correct answer.

Relevance: The KATRIN experiment nearly halved the maximum possible mass for the subatomic particles

Q55.

Answer: b

Explanation:

- **The term 'Blue Washing'**, increasingly seen in environmental news and sustainability debates, refers to a **misleading practice** where industries or entities portray themselves as **environmentally responsible, particularly in the context of marine or water-related sustainability,** while continuing to engage in polluting practices**.** It is derived from the idea of *whitewashing* and is closely related to *greenwashing*, but is specifically focused on **oceans, water, and Sustainable Development Goal 14 (Life Below Water)**.
- In the Indian context, the term has recently gained attention after **the Central Pollution Control Board (CPCB)** reclassified highly polluting industries like **Waste-to-Energy (WTE) incinerators** into a new **'Blue Category'**—traditionally reserved for cleaner, less polluting activities like biogas and composting.

Therefore, option (b) is the correct answer.

Relevance: Central Pollution Control Board (CPCB) reclassified highly polluting industries like Waste-to-Energy (WTE) incinerators into a new 'Blue Category'

Q56.

Answer: b

Explanation:

- **Tuti Island** is located at the **confluence of the Blue and White Nile rivers** in Khartoum**, Sudan.** The island spans 8 square kilometres and was traditionally dependent on farming and fishing, earning the nickname "Khartoum's garden."
    - **Tuti Island has been devastated by two years of war** between the Sudanese army and the paramilitary Rapid Support Forces (RSF), with residents subjected to violence and looting. When fighting broke out on April 15, 2023, RSF fighters swiftly captured the crescent-shaped island, forcing residents to flee in panic.
    - **Climate change has worsened Tuti Island's vulnerability** by disrupting the Nile's natural flooding cycle, causing flooding, droughts, and desertification.
    - Due to the dual threats of conflict and climate change, the **HEART Project (Heritage Empowered Action for Risk in Tuti)** was initiated to preserve the Taya system and document community knowledge. **So, row 1 is correctly matched.**
- **Lake Natron is a salt lake located on the Tanzania-Kenya border, mostly in Tanzania.** It is a Ramsar site and has a unique composition of warm waters and salt, caustic soda, and magnesite deposits that **provide ideal conditions for flamingos to thrive**. But, human activities and changing weather patterns are threatening the world's largest migratory bird colony. **Extreme weather is causing frequent flooding, compromising the lake's salinity.** Also, prolonged dry seasons are also causing water levels to drop dangerously. **So, row 2 is correctly matched.**

- **Santorini** is a popular tourist destination in **Greece's volcanic island**, and has experienced frequent earthquakes recently. **Thousands of people – residents and tourists – had fled Greece's popular tourist destination Santorini after more than 1000 earthquakes** rattled the popular travel destination in a short period. A state of emergency was declared on the Greek island of Santorini after days of consecutive earthquakes. These tremors have been linked to tectonic plate movements from the African-Eurasian Plate interaction. **So, row 3 is not correctly matched.**

So, in two of the above rows is the given information correctly matched.
Therefore, option (b) is the correct answer.

Q57.
Answer: a
Explanation:

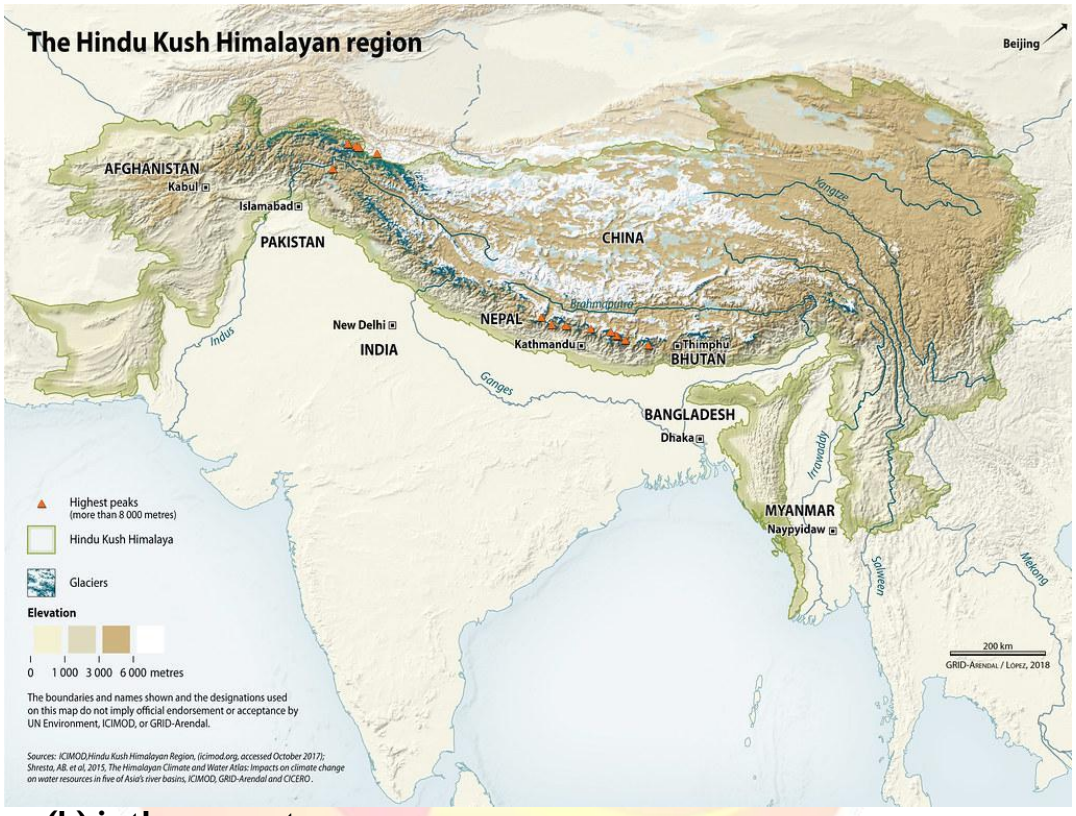
- **Reverse transcriptase loop-mediated isothermal amplification (RT-LAMP) is a rapid, one-step DNA amplification technique**, being projected as an alternative to the widely used RT-PCR (Reverse Transcriptase Polymerase Chain Reaction) technology, which is taken as the current Gold standard for detection of pathogenic genomes like COVID-19 virus in patient's body.
- Though considered gold standard, **RT-PCR technology has certain drawbacks** such as **high turn-around time** - A single test using the technology takes nearly three to four hours to complete, **maintenance of temperature requirement** - DNA amplification through PCR technology is a three step process with every reaction having a distinct temperature requirement and requirement of trained professionals - as **RT-PCR is a technique sensitive process**, trained manpower is required to mann the test.
- RT-LAMP is regarded as a promising point-of-care test due to several features: high sensitivity and specificity, quick reaction time, and reduced dependence on sophisticated equipment.
- **The LAMP reaction takes place at a constant temperature of 60°C to 65°C in under an hour.** Since the DNA amplification process occurs at a constant temperature, RT-LAMP can be performed on simpler equipment and eliminates the need for the thermal cycle in PCR. **So, statement 1 is not correct.**
- A single reaction in RT-LAMP significantly **shortens the reaction time and bypasses the DNA purification step,** thus making it more suitable to use in epidemic situations which require mass screening. **So, statement 3 is correct.**
- RT-LAMP results can be **detected by visual turbidity or fluorescence in real time, while RT-PCR requires blotting and further chromatography for detection.** As such, the results can be interpreted by the untrained naked eye with a short turnaround time, making it **less technique sensitive.** Finally, the simplicity of the procedure means that it can easily be mastered by junior laboratory technicians or health-care workers with little training. **So, statement 2 is not correct.**

So, only one of the above given statements is correct.
Therefore, option (a) is the correct answer.
Relevance: Researchers have developed and tested a novel, cost-effective, real-time LAMP (rt-LAMP) assay for early diagnosis of TB.

Q58.
Answer: b
Explanation:

- **The Hindu Kush Himalaya (HKH)** Mountains extend around 3,500 km. It spreads o**ver eight countries — Afghanistan, Bangladesh, Bhutan, China, India, Nepal, Myanmar, and Pakistan. So, points 1, 2 and 5 are correct.**

- It covers an area of approximately 4.2 million sq.km. The range has numerous high snow-capped peaks, with the highest point being Tirich Mir or Terichmir at 7,708 meters (25,289 ft) in Chitral, Pakistan.
- It is considered the **Third Pole (after the North and South Poles)** and has significant implications for climate. The HKH forms the **largest area of permanent ice cover outside of the North and South Poles** and is home to 4 global biodiversity hotspots.
- The HKH region is the source of ten large Asian river systems: the Amu Darya, Indus, Ganges, Brahmaputra, Irrawaddy, Salween, Mekong, Yangtse, Yellow River, and Tarim.



Therefore, option (b) is the correct answer.

Relevance: The Hindu Kush Himalaya (HKH) region recorded its lowest snow persistence in 23 years during the 2024–2025 winter, according to a new report published recently.

Q59

Answer: a

Explanation:

- Many volcanoes around the world exhibit explosive eruptions. How explosive each eruption is can be measured on a scale known as the Volcanic Explosivity Index.
- The Volcanic Explosivity Index runs from zero to eight, from non-explosive to colossal explosive eruptions. Where a volcano appears on the scale is linked to the amount of material an eruption emits and how frequently eruptions occur.
- A volcano is categorised as **a supervolcano if it can produce a magnitude-eight eruption on the Volcanic Explosivity Index, discharging more than 1,000 cubic kilometres of material.** These eruptions are sometimes called 'super eruptions' and are the **biggest and most explosive of all. So, statement 1 is correct.**

- Super eruptions can produce giant calderas that can be more than 50 kilometres wide. Eruptions of this magnitude are extremely rare. The general trend along the Volcanic Explosivity Index is that the larger the eruption, the less frequently it occurs.
- But **supervolcanoes don't always produce gigantic eruptions**, **sometimes they produce smaller, more 'normal' eruptions**. Naples in Italy might be best known for Mount Vesuvius, but the region is also home to the supervolcano Campi Flegrei – one of the closest supervolcanoes to the UK. These days it's famous for smelly fumaroles and emitting sulphuric acid.
- A magnitude-eight eruption is almost unimaginable. No one alive today has ever seen a supervolcano erupt. **The most recent super eruption was of New Zealand's Taupō volcano, which occurred around 26,500 years ago. So, statement 2 is correct.**
- The 1991 eruption of Pinatubo in the Philippines is one of the largest eruptions in living memory. This event only ranked six on the Volcanic Explosivity Index, making it around 100 times smaller than the benchmark for a supervolcano.
- Supervolcanoes can occur **both at tectonic plate boundaries and at hotspots.** For example, **Yellowstone is located over a hotspot**, while Toba in Indonesia is at a convergent plate boundary. **So, statement 3 is not correct.**

Therefore, option (a) is the correct answer.
Relevance: Scientists have warned of a potential massive volcanic eruption, rivaling Mount Tambora's 1815 disaster

Q60.
Answer: c
Explanation:

- **The Isthmus of Perekop connects the Crimean Peninsula to Kherson Oblast in mainland Ukraine.** It is a narrow strip of land that connects the Crimean Peninsula to the mainland of Ukraine. It is located between Karkinitska Bay, in the Black Sea, and Syvash Lake, toward the Sea of Azov.

Therefore, option (c) is the correct answer.
Relevance: Recently, U.S. President Donald Trump suggested formally recognizing Crimea as part of Russia.

Q61.
Answer: b
Explanation:

- **Flue gas is released from the combustion of coal and other products in thermal power plants.**
- The Union Environment Ministry has mandated India's coal-fired plants to install Flue Gas Desulphurisation (FGD) equipment in order to reduce sulphur dioxide (SO2) emissions. These units remove SO2 from exhaust flue gases.
- Flue-gas desulfurization (FGD) is a scrubbing technique that uses an alkaline reagent (typically a sodium- or calcium-based alkaline regent) to remove SO2 from flue gas. The reagent is injected into the flue gas in a spray tower and directly into the duct and absorbed to neutralize and/or oxidize the SO_2.
- Most FGD technologies use an alkali sorbent, such as **limestone (calcium carbonate), quicklime (calcium oxide), hydrated lime (calcium hydroxide), or sometimes sodium and magnesium carbonate and ammonia**, to capture the acidic sulfur compounds from the flue gas. In all cases the alkalis react with the SO_2 in the presence of water to produce a mixture of sulfite and sulfate salts. **So, points 1, 2 and 4 are correct.**
- The most common FGD technology uses a limestone/gypsum wet-scrubbing process.
- **Flue gas released from power plants contains SO_2 and NO_2 (Nitrogen Dioxide). So, point 3 is not correct.**
- **Aldrin is a pesticide applied to soils to kill termites, grasshoppers, corn rootworm, and other insect pests. It is listed under the Stockholm Convention on Persistent Organic Pollutants. So, point 5 is not correct.**

So, only three of the above mentioned chemicals are used as absorbents in FGD.
Therefore, option (b) is the correct answer.
Relevance: A recent study recommended that the Union Environment Ministry should roll back its 2015 policy mandating all of India's 537 coal-fired plants to install a class of equipment called Flue Gas Desulphurisation (FGD)

Q62.
Answer: a
Explanation:

- **The International Maritime Bureau (IMB) is a specialised division of the International Chamber Of Commerce (ICC).** The IMB is a non-profit making organisation, established in 1981. The primary function of the bureau is to act as a focal point in the fight against all types of maritime crime and malpractice**. So, statement 1 is not correct.**
- **The International Seabed Authority (ISA) is the designated authority to issue licenses for deep-sea mining over the ocean floor beyond national jurisdiction**. Under UNCLOS, exploration for and exploitation of seabed minerals in the 'Area' may only be carried out under a contract with the International Seabed Authority and subject to its rules, regulations and procedures. **So, statement 2 is not correct.**
- The IMB Piracy Reporting Centre is based in Kuala Lumpur, Malaysia. It maintains a round-the-clock watch on the world's shipping lanes, reporting pirate attacks to local law enforcement and issuing warnings about piracy hotspots to shipping.

- IMB PRC offers a 24-hour and free service for shipmasters to report any piracy, armed robbery or stowaway incidents. PRC acts as a single point of contact for **shipmasters anywhere in the world whose vessels have been attacked or robbed by pirates.**
- The key services of the PRC are:
    - Issuing daily status reports on piracy and armed robbery to ships via broadcasts on the Inmarsat-C SafetyNET service.
    - Reporting piracy and armed robbery at sea incidents to law enforcement and the International Maritime Organization (IMO).
    - Helping local law enforcement apprehend pirates and assist in bringing them to justice.
    - Assisting shipowners whose vessels have been attacked or hijacked.
    - Assisting crew members whose vessels have been attacked.
    - Providing updates on pirate activity via the Internet, twitter, and email alerts.
    - Publishing comprehensive quarterly and annual reports detailing piracy statistics.
- The IMB has a MOU with the World Customs Organization (WCO) and **has observer status with Interpol (ICPO). So, statement 3 is correct.**

So, only one of the above given statements is correct.
Therefore, option (a) is the correct answer.
Relevance: Global Maritime piracy and armed robbery incidents have seen a noticeable increase in the first three months of 2025, according to the International Maritime Bureau (IMB).

Q63
Answer: a
Explanation:

- **Mist nets are rectangular nets composed of fine threads. Mist nets are a tool for bird population monitoring.** Mist nets are an important tool for population monitoring including assessment of species composition, relative abundance, population size, and demography. Mist netting is a survey method to gain valuable information about bird and bat populations.
- Advantages of using mist nets over aural or visual count methods include ease of standardized sampling, low observer bias, ability to detect species that are often missed using other count methods, and opportunity to examine birds in the hand.

Therefore, option (a) is the correct answer.

Q64.
Answer: a
Explanation:

- **The Sillahalla Pumped Storage Hydroelectric Project** (SPSHEP) is a major infrastructure initiative proposed **by the Tamil Nadu** Green Energy Corporation Limited (TNGECL) to address the state's peak power demand and support renewable energy integration. It will be **located in Nilgiris district, of Tamil Nadu.** The Sillahalla Pumped Storage Hydro-Electric Project aims to construct two reservoirs in the Kundah region of Tamil Nadu's Nilgiri district**. So, pair 1 is correctly matched.**
- **Balpakram National Park** is situated in the West Garo Hills district of the north-eastern state of **Meghalaya.** It is known as the "Land of Perpetual Winds" due to strong winds that blow across the

plateau. It is also close to the international boundary of Bangladesh. It was declared a National Park in 1987. **So, pair 2 is not correctly matched.**

- **Thangjing Hill is a mountain peak located in Manipur.** It is in the Churachandpur district, to the west of Moirang. The north-south-running mountain range on which it sits is also called Thangjing range or Thangjing Hills. **So, pair 3 is not correctly matched.**
    - It is believed by Manipuris to be the abode of Thangching, the ancestral deity of Moirang. It is considered a sacred site by the Meitei community, who traditionally visit it throughout the Manipur month of 'Sajibu' which generally falls in April.

Therefore, option (a) is the correct answer.
Relevance: Tension prevailed near the buffer zone in Manipur's Thangjing Hill range recently, which separates Kuki and Meitei inhabited areas in Churachandpur district

Q65.
Answer: b
Explanation:

- **MacGregor Memorial Medal** was instituted on 03 July 1888 to commemorate the memory of Maj. Gen. Sir Charles Metcalfe MacGregor, the founder of the United Service Institution of India (USI). This medal is granted annually by the **United Service Institution of India (USI). So, statement 1 is not correct.**
    - United Service Institution of India (USI) is a national security and defence services think tank based in New Delhi, India. It was founded in Simla in 1870 by Col. (later Maj Gen) Charles Metcalf MacGregor, to advance military knowledge and professional development among British Indian Army officers. It is the oldest autonomous think tank in India.
- Originally intended **to recognise the acts of military reconnaissance and exploratory journeys,** the medal's scope was expanded in 1986, post-independence, to include **military expeditions and adventure activities. So, statement 2 is correct.**
- The medal is **open to all ranks, serving and retired, of the Indian Armed Forces, Territorial Army, and Assam Rifles.** This medal is awarded to Armed Forces personnel for valuable contributions in military reconnaissance, expeditions and adventure activities.

Therefore, option (b) is the correct answer.
Relevance: Five military personnel were recently conferred the MacGregor Memorial Medal

Q66.
Answer: d
Explanation:

- Type 5 diabetes h**as been recently officially recognised as a distinct form of the disease by the International Diabetes Federation (IDF).** It was officially recognised at the 75th World Congress for Diabetes in Bangkok.
- Type 5 diabetes is a form of diabetes affecting lean and malnourished teenagers and young adults in low- and middle-income countries.
- Type 5 diabetes **affects lean and undernourished young adults in low- and middle-income countries.** It is **caused by malnutrition-**induced lowering of insulin production. It currently affects an estimated 25 million people globally. **So, statement 2 is correct.**
- This unique form of diabetes has no evidence of autoimmune or genetic causes.

- The disease **causes pancreatic beta cells to function abnormally, leading to insufficient production of insulin. So, statement 1 is not correct.**
    - This is unlike Type 2 diabetes, the most prevalent form of the disease, where the main problem is insulin resistance. Here, the pancreas continues to produce insulin, but the body does not respond properly to the hormone.

So, Statement–I is incorrect, but Statement–II is correct.
Therefore, option (d) is the correct answer.
Relevance: Neglected for decades in research, and frequently misdiagnosed, it has taken a long time for Type 5 diabetes, caused by malnutrition-induced lowering of insulin production, to gain official recognition.

Q67.
Answer: c
Explanation:

- BharatNet is an ambitious project of the Government of India aimed at providing broadband connectivity to all Gram Panchayats (GPs) in the country. It is one of the biggest rural telecom projects in the world. The next stage of the project, the **Amended BharatNet Program (ABP)** was approved in August 2023.
- The ABP is a design improvement **aiming for Optical Fibre (OF) connectivity** to 2.64 lakh **GPs in ring topology (a network design where connected devices form a circular data channel)** and of **connectivity to the remaining non-GP villages on demand. So, statement 1 is not correct but 2 is correct.**
- ABP includes features like IP-MPLS (Internet Protocol Multi-Protocol Label Switching) network with routers at Blocks and GPs, operation and maintenance for 10 years, power backup, and Remote Fibre Monitoring System (RFMS). The cost allocated is Rs. 1,39,579 crores.
- The BharatNet project is being executed by a Special Purpose Vehicle (SPV) namely Bharat Broadband Network Limited (BBNL). Under the Amended BharatNet Program, **BSNL is appointed as the single Project Management Agency (PMA) for Operation & Maintenance of the entire network. So, statement 3 is correct.**

Therefore, option (c) is the correct answer.

Q68.
Answer: d
Explanation:

- **I**ndia's Direct Benefit Transfer (DBT) system has helped the country achieve **cumulative savings of ₹3.48 lakh crore by plugging leakages in welfare delivery**, according to a new quantitative assessment by the BlueKraft Digital Foundation.
- The report also finds that **subsidy allocations have been halved from 16 percent to 9 percent of total government expenditure** since the implementation of DBT, reflecting a major improvement in the efficiency of public spending. **So, point 1 is correct.**
    - Pre-DBT Era (2009–2013): Subsidies averaged 16% of total expenditure, amounting to ₹2.1 lakh crore annually, with considerable leakages in the system.
    - Post-DBT Era (2014–2024): Subsidy expenditure decreased to 9% of total expenditure in 2023-24, **while beneficiary coverage surged 16-fold from 11 crore to 176 crore. So, point 2 is correct.**

- The report highlights that there is a strong positive correlation between beneficiary coverage and DBT savings, signifying that as coverage expanded, savings increased.
- As per the report, as DBT savings increased, subsidy allocations decreased, demonstrating that DBT improved targeting while reducing leakages. This enabled the **government to expand welfare programs, reaching more beneficiaries without increasing fiscal outlays. So, point 3 is correct.**

Therefore, option (d) is the correct answer.

Relevance: According to a new quantitative assessment by the BlueKraft Digital Foundation, India's Direct Benefit Transfer (DBT) system has helped the country achieve cumulative savings of ₹3.48 lakh crore by plugging leakages in welfare delivery

Q69.

Answer: a

Explanation:

- **Taj Trapezium Zone (TTZ) is a defined area of 10,400 sq km around the Taj Mahal to protect the monument from pollution**. **So, statement 1 is correct.**
- Taj Trapezium Zone (TTZ) **covers the districts of Agra, Firozabad, Mathura, Hathras, and Etah in Uttar Pradesh and Rajasthan's Bharatpur** district. **So, statement 2 is not correct.**
- The Supreme Court recently tasked the National Environmental Engineering Research Institute (NEERI) **to study the impact of glass industries on the Taj Mahal,** a UNESCO World Heritage site. The Court further said that in case of an adverse report from NEERI, it would not hesitate to pass a direction for the removal of the industrial units. **So, statement 3 is not correct.**
    - The Court has banned the use of coal/ coke in industries located in the TTZ. Only environmentally friendly, non-polluting small, tiny and micro scale industries may be allowed to operate in this zone.
- The Supreme Court recently ordered a tree census in an area spread across 10,400 kilometres around the iconic Taj Mahal in Agra, known as Taj Trapezium Zone (TTZ).
    - The Court has said that if trees are felled without permission, there are penal provisions under the UP Protection of Trees Act, 1976. Penal provisions cannot be implemented unless data of trees is available. That data can be available only if a census is available. Without census, there cannot be proper implementation of the 1976 Act.
- There is **not blanket ban on felling of trees** within the Taj Trapezium Zone. The Supreme Court has recently mandated **its prior permission for Tree Felling within 5 Kms of Taj Mahal. So, statement 4 is not correct.**
- According to Court, tree felling in areas within TTZ beyond the 5 kilometer distance of the historical monument will require the prior permission of the divisional forest officer (DFO) of the central empowered committee (CEC) and the officer will be bound by the provisions of UP Tree Preservation Act.

Therefore, option (a) is the correct answer.

Relevance: The Supreme Court recently tasked the National Environmental Engineering Research Institute (NEERI) to study the impact of glass industries on the Taj Mahal

Q70.

Answer: b

Explanation:

- Svalbard archipelago is a group of islands north of continental Norway. It is located in the Arctic Ocean approximately 650 miles from the North Pole.

- In 1920, the Svalbard Treaty was signed. **The Svalbard Treaty granted Norway sovereignty over the archipelago.** The Svalbard Treaty recognises the sovereignty of Norway over the Arctic archipelago of Svalbard. **So, statement 1 is not correct.**
    - The **Arctic Council** was established on 19 September 1996 when the governments of Canada, the Kingdom of Denmark, Finland, Iceland, Norway, the Russian Federation, Sweden, and the United States signed the **Ottawa Declaration.**
- Prior to 1920 Svalbard was considered terra nullius and open for all states and stakeholders who wanted to take advantage of the area's rich resources.
- However, stakeholders from signatory states **enjoy the same rights of access to Svalbard as does Norway,** when it comes to exploitation of natural resources. **So, statement 2 is correct.**
    - The treaty obliges Norway to grant citizens and commercial entities from the co-signing nations equal access within certain specific contexts:
        - access to and presence in Svalbard
        - fishing and hunting
        - all types of maritime, industrial, mining, and commercial operations
        - acquisition, exercise, and enjoyment of the right to ownership of property, including mineral rights
- The Svalbard Treaty states that Svalbard cannot be used for war purposes and that Norway cannot have military bases there.
- As of 2024, there are 48 parties to the treaty. **India is a party to the treaty.**

Therefore, option (b) is the correct answer.

Q71.
Answer: c
Explanation:

- The **prevailing interest rates generally have an inverse relationship with domestic gold prices.** If **the interest rates increase**, there is a heavy sell-off of gold, increasing supply. **This puts downward pressure on gold prices**. People prefer to buy gold when the interest rates decrease, increasing the demand. It is often seen that the prices of gold increase after a reduction in the interest rates. Gold tends to be a preferred investment when the interest rates drop. Gold typically benefits from lower interest rates as it reduces the opportunity cost of holding the non-yielding asset like gold. Lower interest rates reduce the opportunity cost of holding gold. Thus, the demand for gold increases and so the prices. **So, point 1 is not correct.**
- **In recent times, the American Dollar has slumped sharply.** Waning confidence in the American economy is leading to a huge flight out of the greenback to other safe haven currencies and gold. Recent fluctuations in the value of the US dollar have prompted businesses to invest in gold to hedge their currency assets. The dollar sank to a fresh two year low. **A weaker dollar boosts demand for gold, putting upward pressure on price. So, point 2 is correct.**
- **Recently, there has been a notable trend of central banks around the world accumulating large quantities of gold**. Analysts say the yellow metal is likely to stay strong in 2025 amid global uncertainty and sustained demand from central banks. According to the World Gold Council, central banks globally added a net 44 tonnes of gold to their reserves in January and February 2025.

- When central banks buy large quantities of gold, it signals increased demand for the metal, which directly impacts its price. **The more central banks hold in reserves, the tighter the supply becomes, leading to upward pressure on prices. So, point 3 is correct.**
- **Global concerns on inflation in 2025 on the back of various factors such as tariffs, a weaker dollar, and supply constraints has also been a driver.** Experts note that gold acts as a hedge against rising inflation. Gold prices react to inflation. When inflation rises the currency values go down. Thus, people tend to hold money in the form of gold. Gold acts as a hedging tool against inflationary conditions. **So, point 4 is not correct.**

Therefore, option (c) is the correct answer.

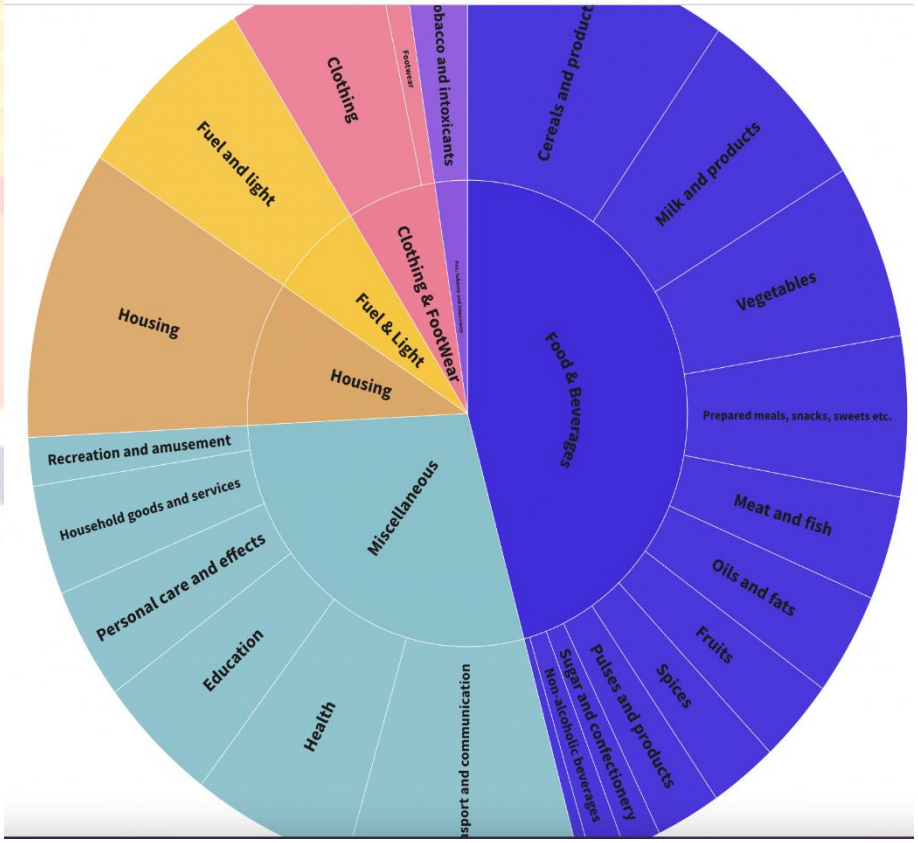
Relevance: Gold prices have crossed ₹1 lakh per 10 grams for the first time, driven by a surge in international markets after US President Donald Trump announced plans to overhaul the Federal Reserve.

Q72.

Answer: b

Explanation:

- Consumer Price Index (CPI) inflation tracks the change in retail prices of goods and services which households purchase for their daily consumption. CPI is calculated (base year is 2012) for a fixed basket of goods and services. The National Statistical Office (NSO), Ministry of Statistics and Programme Implementation (MoSPI), compiles CPI for All India and States for Rural, Urban, Combined sectors and releases it every month. The RBI currently uses CPI as a key measure of inflation to set monetary and credit policy.
    - Food and Beverages has a weightage of 45.86%.
    - Housing has a weightage of 10.07%
    - Health and Education has a combined weightage of 10.35%
    - Fuel and Light has a weightage of 6.84%
    - Clothing and Footwear has a weightage of 6.53%



- So, the correct increasing order is **Clothing and Footwear < Fuel and Light < Housing < Health and Education < Food and Beverages**

Therefore, option (b) is the correct answer.

Q73.
Answer: b
Explanation:

- **The International Big Cat Alliance (IBCA)** signed an agreement with the central government recently **formalising India as the headquarters and secretariat of the alliance. So, point 1 is correct.**
    - The IBCA was launched at the initiative of India to focus on global conservation of seven big cats —the tiger, lion, leopard, snow leopard, puma, jaguar and cheetah.
- The **Mangrove Alliance for Climate** is an intergovernmental alliance that was launched at the 27th Session of Conference of Parties (COP27). It seeks to educate and spread awareness worldwide on the role of mangroves in curbing global warming and its potential as a solution for climate change. An initiative led by the United Arab Emirates (UAE) and Indonesia, the Mangrove Alliance for Climate (MAC) also includes India, Sri Lanka, Australia, Japan, and Spain. **Its headquarter is in Dubai, United Arab Emirates. So, point 2 is not correct.**
- **The International Solar Alliance** is a global intergovernmental organization dedicated to advancing solar power adoption for a carbon-neutral future. It was conceptualized on the side lines of COP21 in Paris in 2015. Its mission is to unlock $1 trillion in solar investments by 2030 while reducing technology and financing costs. **Its Secretariat is in Gurugram, India. So, point 3 is correct.**
- **The Coalition for Disaster Resilient Infrastructure (CDRI)** is a multi-stakeholder global partnership of national governments, UN agencies, multilateral development banks and the private sector. The Prime Minister of India launched CDRI during his speech at the UN Climate Action Summit on 23 September 2019. Its secretariat is at **New Delhi (India)**. **So, point 4 is correct.**
- **The International Tropical Timber Organization (ITTO)** is an intergovernmental organization which aims to promote sustainable management and conservation of tropical forests. Its **headquarters are located in Yokohama, Japan**. **So, point 5 is not correct.**

So, the headquarters/secretariat of only three of the above are located in India.
Therefore, option (b) is the correct answer.
Relevance: India and International Big Cat Alliance sign agreement to host its headquarters

Q74.
Answer: d
Explanation:

- In a step toward sustainable and environmentally friendly mining practices, South Eastern Coalfields or SECL is set to become the first coal public sector undertaking (PSU) in India to adopt **paste fill technology for coal mining.** To implement this underground mining technology, SECL has signed a Rs 7,040 crore agreement with TMC Mineral Resources.
- **Paste fill technology is a modern underground mining method** that eliminates the need to acquire surface land. After coal extraction, the mined-out voids are filled with a specially prepared paste made from fly ash, crushed overburden from opencast mines, cement, water and binding chemicals. This process prevents land subsidence and ensures the structural stability of the mine.

Therefore, option (d) is the correct answer.
Relevance: South Eastern Coalfields Limited (SECL) is set to become the first coal PSU in India to adopt paste fill technology for coal mining—marking a major step toward sustainable and environmentally friendly mining practices.

Q75.
Answer: c
Explanation:

- The **Foreign Contribution (Regulation) Act, 2010** has come into effect from May 1, 2011. The Act regulates the acceptance and utilisation of foreign contribution by individuals, associations and companies.
- The FCRA requires every person or NGO seeking to receive foreign donations to be (i) registered under the Act, (ii) to open a bank account for the receipt of the foreign funds in State Bank of India, Delhi, and (iii) to utilise those funds only for the purpose for which they have been received and as stipulated in the Act.
- Under the Act, certain persons are prohibited to accept any foreign contribution. These include: election candidates, editor or publisher of a newspaper, **judges, government servants,** public servants, members of any legislature, and political parties**. So, statement 1 is correct.**
- Under the Act, **a person who receives foreign contribution must use it only for the purpose for which the contribution is received**. Further, they must not use more than 50% of the contribution for meeting administrative expenses. **So, statement 2 is correct.**
- The Act **prohibits the transfer of foreign contribution to any other person. So, statement 3 is correct.**
- **Contributions made by a citizen of India living in another country** (i.e., Non-Resident Indian), from his personal savings, through the normal banking channels, **are not treated as foreign contributions. So, statement 4 is not correct.**

Therefore, option (c) is the correct answer.

Q76.
Answer: c
Explanation:

- Vehicle-to-Grid (V2G) refers to technologies that **enable Electric Vehicle batteries to send power back to the power grid.** When an EV is not in use, it can act as a decentralised battery energy storage device. An idle EV, when connected to a bi-directional charger, can provide support to the distribution grid. By incorporating V2G technologies, EVs offer an opportunity to facilitate the integration of Renewable Energy (RE) and support a demand response market. **So, statement 1 is correct.**
- With V2G technologies, it is **possible for EVs to supply electricity to the grid during times of high demand**. This will help in improving **grid stability. So, statement 2 is correct.**
- The **Kerala State Electricity Board (KSEB)** and the Indian Institute of Technology Bombay (IIT Bombay) have **initiated a pilot project** to explore the implementation of **V2G technology across the State.** This collaboration aims to assess the feasibility of integrating EVs into the State's power grid. The project plans to assess an EV's ability to support the grid during peak demand when solar energy is not available. **So, statement 3 is not correct.**

Therefore, option (c) is the correct answer.
Relevance: The Kerala State Electricity Board (KSEB) and the Indian Institute of Technology Bombay (IIT Bombay) have initiated a pilot project to explore the implementation of Vehicle-to-Grid (V2G) technology.

Q77.
Answer: d
Explanation:

- Taurus is a modular, air-launched, long-range, precision-guided **cruise missile**. It has a range of **more than 500km**. **So, statement 1 is not correct.**

- The Taurus missile system has been developed **jointly by German defence firm MBDA and Sweden's Saab. So, statement 2 is not correct.**
- Taurus missiles are equipped with advanced warheads designed to cause heavy damage to fortified infrastructure like bridges and bunkers.

Therefore, option (d) is the correct answer.
Relevance: Recently, Russia had apprehensively responded to Germany's proposal to the use of Taurus Missiles in Russsia-Ukraine war.

Q78.
Answer: a
Explanation:

- The Dawoodi Bohras originated from Western India. **They constitute a Muslim community** with a global presence across more than 40 nations. It is a sect of the Muslim. **The Dawoodi Bohras are Shia Muslims. So, statement 1 is not correct.**
- Their ancestral lineage connects to the Fatimid Imams of Egypt, who were direct descendants of Prophet Muhammad.
- **The community follows the leadership of al-dai al-mutlaq** (unrestricted missionary), who initially operated from Yemen before establishing a base in India 450 years ago. **So, statement 2 is correct.**
- For about 450 years, **the leader of the community has been based in India**, including the current and the 53rd leader, His Holiness Dr Syedna Mufaddal Saifuddin. **So, statement 3 is not correct.**

Therefore, option (a) is the correct answer.
Relevance: A Dawoodi Bohra community delegation met Prime Minister Narendra Modi and thanked him for the newly passed Waqf (Amendment) Act, 2025

Q79.
Answer: b
Explanation:

- **The Mission Amrit Sarovar** was announced on 24th April 2022 by the Prime Minister during the celebration of National Panchayati Raj Day.
- The Mission Amrit Sarovar **was launched as a flagship initiative under the Azadi Ka Amrit Mahotsav**. **So, statement 1 is correct.**
- The mission **aims to construct and rejuvenate 75 water bodies in each district across the country, thereby fostering water conservation,** ensuring sustainability, and reviving traditional community water bodies through public participation. **So, statement 2 is not correct.**
- This mission is supported by the Bhaskaracharya National Institute for Space Applications and Geo-informatics (BISAG-N). This multi-stakeholder approach was intended to ensure convergence, efficiency, and participatory ownership of the initiative. **The National Informatics Centre (NIC) has provided a centralized digital platform—amritsarovar.gov.in—which tracks real-time progress** at the granular level, ensuring transparency and enabling coordination between departments and states. **So, statement 3 is correct.**
- **The Mission is anchored by the Ministry of Rural Development.** It involves seven ministries: Ministry of Rural Development, Ministry of Jal Shakti, Ministry of Panchayati Raj, Ministry of

Environment, Forest & Climate Change, Ministry of Railways, Ministry of Culture and Ministry of Road Transport and Highways. **So, statement 4 is not correct.**

- Mission Amrit Sarovar works are being taken up by the states and districts with convergence from various ongoing schemes such as Mahatma Gandhi National Rural Employment Guarantee Scheme (Mahatma Gandhi NREGS), 15th Finance Commission Grants, Pradhan Mantri Krishi Sichayi Yojna sub-schemes etc.

So, only two of the above statements are correct.
Therefore, option (b) is the correct answer.
Relevance: As of March 2025, over 68,000 Sarovars have been completed, enhancing surface and groundwater availability across various regions.

Q80.
Answer: d
Explanation:

- The Supreme Court of India has upheld the **right of homebuyers to peacefully protest against builders**, saying portraying their **right to free speech** as a criminal offence is a clear abuse of the law.
- The apex court said that the right of homebuyers to peacefully protest against builders is protected under t**he right to free speech and expression enshrined in Article 19** of the Constitution.

Therefore, option (d) is the correct answer.
Relevance: The Supreme Court has upheld the right of homebuyers to peacefully protest against builders, saying portraying their right to free speech as a criminal offence is a clear abuse of the law.

Q81.
Answer: a
Explanation:

- A **cloudburst** is a sudden spell of localised, heavy rainfall where the amount of rainfall over a particular area exceeds 100 millimetres in an hour. They often result in flash floods and have become increasingly common from May-September when the southwest monsoon season prevails in much of the country.
- **The phenomenon is more common in mountainous areas** because they offer the terrain for moisture-laden air to rise swiftly along the mountain slopes. **So, statement 1 is correct.**
- The phenomenon that is responsible for this is **'orographic lift', a process by which clouds that are already to rain are pushed up by warm air currents**. As they reach higher elevations, the water droplets within the clouds become bigger and new ones are formed. **So, statement 2 is correct.**
- These dense clouds eventually burst, unable to hold the large volume of moisture. This results in torrential downpours in the region right below and leads to overflowing of water bodies in a very short duration of time.

So, both Statement–I and Statement–II are correct and Statement–II explains Statement–I.
Therefore, option (a) is the correct answer.

Q82.
Answer: b
Explanation:

- **Two cheetahs, Prabhas and Pawak, recently moved** to their new home as they were released into the **Gandhi Sagar Wildlife Sanctuary.** The cheetahs were moved from the Kuno National Park, which until now was the only home for the big cats in the country since their translocation from

Africa in September, 2022. Four cheetahs are scheduled to be translocated to Gandhi Sagar from Botswana in May. **So, pair 1 is correctly matched.**

- **Asola Bhatti Wildlife Sanctuary** is located in the South Delhi Ridge of the Aravalli hill range on the Delhi-Haryana border. The sanctuary is part of the Southern Ridge and has biodiversity significance as it merges with the Indo-Gangetic Plain. It is an important part of the North Aravalli Leopard Wildlife Corridor,which starts from Sariska National Park in Rajasthan, passes through various districts of Haryana, and finally meets the Delhi Ridge.
    - **Madhav National Park i**n Shivpuri district of Madhya Pradesh, was declared a tiger reserve recently, taking the tally of such reserves in the country to 58. **So, pair 2 is not correctly matched.**
- **Bhadra Wildlife Sanctuary is in Karnataka.** It is named after the Bhadra River, which flows through the sanctuary. It was declared a Tiger Reserve in 1998. Karnataka Forest Department will **'soft release' captured Elephants** into Bhadra sanctuary to mitigate human-elephant conflict. **So, pair 3 is correctly matched.**
    - Soft-Release Strategy: Elephants will stay in a 20 sq. km enclosure for acclimatization and health checks before their phased release at four designated sites in BWS.

Therefore, option (b) is the correct answer.

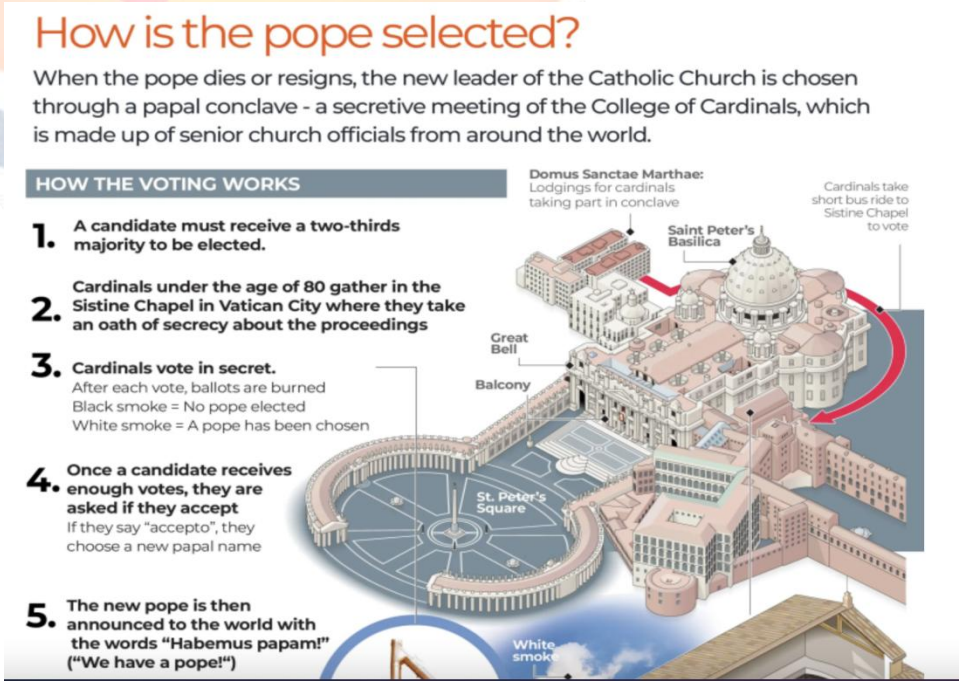
Relevance: Two cheetahs, Prabhas and Pawak, recently moved to their new home as they were released into the Gandhi Sagar Wildlife Sanctuary.

Q83.

Answer: c

Explanation:

- The **Pope** is the head of the **Catholic Church**, and is an important source of authority for the world's roughly 1.4 billion Roman Catholics.
- The College of Cardinals, composed of senior Catholic clergy, elects the next Pope. The new Pope is **elected by the cardinals - all men who were appointed by the Pope and are usually ordained bishops**. The Conclave elects the new pope. **Cardinals below the age of 80,** when the pope dies or resigns, vote in what is known as the Papal Conclave. **So, statement 1 is not correct.**
- In theory, **any Roman Catholic man** who has been baptised can be considered to become Pope. In practice, however, the cardinals prefer to select one of their own.
- There are currently 252 Catholic cardinals, but only 135 can cast ballots as **those over the age of 80 can take part in debate but cannot vote. So, statement 2 is correct.**
- Each cardinal votes for the candidate they wish to become Pope. They can vote for themselves if they wish. While the number of electors is typically capped at 120, there are currently 135 eligible voters. Its members cast their votes



via secret ballots, a process overseen by nine randomly selected cardinals. **A two-thirds majority is traditionally required to elect the new pope,** and voting continues until this threshold is met. **So, statement 3 is correct.**

Therefore, option (c) is the correct answer.

Relevance: After the death of Pope Francis, on May 7, cardinals will enter the Sistine Chapel to elect the next head of the Catholic Church.

Q84.

Answer: c

Explanation:

- In warm and tropical climates, excess solar gain may result in high cooling energy consumption. Shading is a simple method to block the sun before it can get into the building. **The 'brise-soleil', or 'sun-breaker' solution refers to a permanent sun shading technique,** like the simple patterned concrete walls popularized by Le Corbusier.
- **Brise soleil** means 'sun breaker'. It refers to **permanent sun-shading techniques** ranging from the simple patterned concrete walls popularized by Le Corbusier to the elaborate wing-like mechanism devised by Santiago Calatrava. The brise-soleil concept is used to **prevent a heavily glazed facade from overheating during the summer.** It is a technique that reduces heat gain by buildings by deflecting incoming sunlight.

Therefore, option (c) is the correct answer.

Q85.

Answer: d

Explanation:

- Rare earth elements are a series of 17 substances that are present in the earth's crust. Unlike what the name may indicate, rare earths occur plentifully in nature, but the rarity comes from the ability to isolate them chemically and make them usable in industrial applications.
- Amid the ongoing U.S.–China trade war, Chinese authorities have reportedly imposed export controls on rare earth elements (REEs). China's curbs target dysprosium, gadolinium, lutetium, samarium, scandium, terbium, and yttrium, seven of the 17 elements classified as REEs.
    - **Dysprosium's so-called magnetic susceptibility makes it ideal for use in hard disks** and car motors. **So, pair 3 is correctly matched.**
    - **Gadolinium is used in nuclear reactors** for shielding and in some medical equipment. Gadolinium is excellent at absorbing neutrons, and so is used in the core of nuclear reactors. **So, pair 1 is correctly matched.**
    - Lutetium and compounds with it are used in PET scanners and in petroleum refineries.
    - Samarium is used in powerful magnets in personal electronics.
    - Scandium–aluminium alloys are used in fighter aircraft.
    - Terbium is used in lighting for personal electronics.
    - **Yttrium is used in cancer treatments and superconductors**. Yttrium is a silvery-metallic transition metal chemically similar to the lanthanides and is classified as a "rare-earth element". Yttrium is used in drugs to treat rheumatoid arthritis and cancer. **So, pair 2 is correctly matched.**

- Scandium is a chemical element with the symbol Sc and atomic number 21. It is often classified as a rare earth element due to its properties and occurrence in nature, and is commonly found in ores of various metals like aluminum, cobalt, and titanium. Scandium is used in televisions and fluorescent lamps.
- **Lanthanum finds application in X-ray films as phosphor.** Yttrium, europium and terbium are also used as phosphors in displays of computers, TV, etc. **So, pair 4 is correctly matched.**

So, all four of the pairs given above are correctly matched.
Therefore, option (d) is the correct answer.
Relevance: Amid the ongoing U.S.–China trade war, Chinese authorities have reportedly imposed export controls on rare earth elements (REEs)

Q86.
Answer: b
Explanation:

- The **run-off factor refers to the percentage of deposits that could be withdrawn by depositors in a stress scenario.** Internet and Mobile Banking facilities (IMB) includes all facilities such as but not limited to internet banking, mobile banking and Unified Payments Interface (UPI) which enables a customer to digitally transfer funds from their accounts.Run-off factor means the possibility of deposits getting withdrawn/transferred, including in stressed situations.
- The Reserve Bank of India (RBI) recently eased the liquidity coverage ratio (LCR) norms under which the banks will now be required to assign an additional 2.5 per cent run-off factor for digital deposits.
- "A bank shall assign an additional 2.5 per cent run-off factor for retail deposits which are enabled with internet and mobile banking facilities (IMB)," the RBI said.

Therefore, option (b) is the correct answer.
Relevance: RBI mandates 2.5% additional run-off factor on digital deposits

Q87.
Answer: c
Explanation:

- **International Telecommunication Union (ITU) is headquartered in Geneva,** Switzerland, and with regional offices on every continent. **It is the oldest agency in the UN family** – connecting the world since the dawn of the telegraph in 1865. **So, statement 1 is correct**.
- ITU is the United Nations specialized agency for digital technologies (ICTs). **The Organization is made up of a membership of 194 Member States and more than 1000 companies, universities and international and regional organizations. So, statement 2 is not correct.**
- ITU coordinates the global telecom systems through its three sectors: ITU-T (Standardization), ITU-D (Development), and ITU-R (Radiocommunication).
    - The Radiocommunication Bureau manages the **global radio-frequency spectrum and satellite orbit resources,** ensuring countries and technologies (like 5G, aviation, space missions, etc.) don't interfere with one another. This coordination is crucial for everything from mobile networks to GPS, weather satellites, and broadcasting. **So, statement 3 is correct.**

Therefore, option (c) is the correct answer.
Relevance: India has nominated Ms. M. Revathi, Joint Wireless Advisor at the Department of Telecommunications (DoT), as its candidate for Director of the Radiocommunication Bureau at the International Telecommunication Union (ITU).

Q88.
Answer: a
Explanation:

- The Simla agreement was signed on July 2, 1972, after three days of talks between Prime Minister Indira Gandhi, victorious from the 1971 Bangladesh War, and Pakistan Prime Minister Zulfikar Ali Bhutto, who had suffered a crushing defeat and the division of Pakistan in the war.
- **In the Simla agreement, both India and Pakistan pledged to resolve all disputes—including the Kashmir issue—bilaterally and without third-party intervention**. This was a major shift from earlier stances that had invited or accepted international mediation. **So, statement 1 is correct.**
- In the case of India and Pakistan, the **"Ceasefire Line" was established by the Karachi Agreement (1949) after the 1948-49 war.** The Simla Agreement re-characterised this line as the **"Line of Control", a military border without "prejudice to the recognised position of either side"** – which means this is not a legally recognised international border but a holding line for both militaries. **So, statement 2 is not correct**.
- As part of the Simla agreement, India agreed to **release over 93,000 Pakistani prisoners of war** — one of the largest post-war prisoner releases in history. **So, statement 3 is correct.**
- The Agreement saw the **first formal recognition of Bangladesh** as a sovereign entity in a bilateral treaty between India and Pakistan. **So, statement 4 is not correct.**

Therefore, option (a) is the correct answer.

Relevance: Pakistan's National Security Committee (NSC) issued a threat to suspend its participation in all bilateral agreements with India, including the Simla Agreement, after the Pahalgam incident.

Q89.
Answer: d
Explanation:

- **The India-China border** is divided into three sectors, viz. Western, Middle and Eastern. The boundary dispute in the **Western Sector pertains to the Johnson Line** proposed by the British in the 1860s that extended up to the **Kunlun Mountains and put Aksai Chin in the then princely state of Jammu and Kashmir**. Independent India used the Johnson Line and claimed Aksai Chin as its own. **So, pair 1 is not correctly matched.**
- The disputed boundary in the **Eastern Sector of the India-China border is over the McMahon Line**. Representatives of China, India, and Tibet in 1913-14 met in Shimla, where an agreement was proposed to settle the boundary between Tibet and India and Tibet and China. Though the Chinese representatives at the meeting initialled the agreement, they subsequently refused to accept it. **So, pair 2 is correctly matched.**
- **The Radcliffe Line is the boundary that was drawn to demarcate the border between India and Pakistan after the Partition of India in 1947**. It was named after Sir Cyril Radcliffe, the chairman of the Boundary Commission. **So, pair 3 is correctly matched.**
- **The Durand Line is an international land border that stretches for about 2,600 km between Afghanistan and Pakistan**. It culminates at its western end at the border with Iran and at its eastern end at the border with China. It was established in 1893 as the border between British India and the Emirate of Afghanistan, marking their respective spheres of influence. **So, pair 4 is not correctly matched.**

- **The Iran-Pakistan border, known as the Goldsmith Line,** stretches from a tripoint with Afghanistan to the northern Arabian Sea. **So, pair 5 is correctly matched.**

Therefore, option (d) is the correct answer.

Relevance: Pakistani troops violated the ceasefire along several sectors at the Line of Control (LoC) and International Border after the Pahalgam incident.

Q90.

Answer: d

Explanation:

- The **Jalaj initiative under the aegis of National Mission for Clean Ganga (NMCG)**, is a significant component of the Government's Arth Ganga vision — linking people to rivers through sustainable economic activities. **JALAJ aims at creating economic opportunities and enhancing community participation in river rejuvenation efforts**.
- Jalaj has successfully connected river conservation with livelihood generation by promoting eco-tourism, sustainable farming, skill development, and artisanal production across the Ganga basin. The initiative aims to establish 75 Jalaj centers, with various models such as Dolphin Safaris, Homestays, Livelihood Centers, and Awareness & Sale Points.

Therefore, option (d) is the correct answer.

Relevance: The Union Minister of Jal Shakti chaired a review meeting to assess the progress of the JALAJ.

Q91.

Answer: a

Explanation:

- **Silkyara tunnel** is a 4.531 km long, two-lane bi-directional tunnel with an escape passage, located on the **Dharasu-Yamunotri section in Uttarakhand.** The Silkyara Tunnel is part of the infrastructure developed to support the Char Dham Yatra in Uttarakhand and was built using the New Austrian Tunnelling Method. **So, pair 1 is correctly matched.**
- **Z-Morh tunnel is a 2-lane road tunnel between Gagangir and Sonamarg on the Srinagar-Leh Highway in Jammu and Kashmir**. The name "Z-Morh" refers to the Z-shaped road section near the construction site. In addition to providing all-weather access to Sonamarg, the tunnel is vital for ensuring year-round connectivity to Ladakh. This is particularly important for the movement of military personnel to border areas. **So, pair 2 is not correctly matched.**
- **The Sela tunnel** has been constructed by the Border Roads Organisation (BRO) at an altitude of 13,000 feet on the road connecting **Tezpur in Assam, to Tawang in the West Kameng district of Arunachal Pradesh**. Built at a total cost of Rs 825 crore, the tunnel will provide all-weather connectivity to Tawang across Sela Pass on the Balipara - Chariduar - Tawang Road, boosting the preparedness of the Armed Forces and augmenting the socio-economic development of the border region. **So, pair 3 is not correctly matched.**

Therefore, option (a) is the correct answer.

Relevance: Prime Minister Narendra Modi inaugurated the Z-Morh tunnel in January 2025.

Q92.
Answer: a
Explanation:

- There are **five species of rhinos and these include two African rhino species** - black and white rhinos. The remaining three are Asian rhino species, which include greater one–horned (Indian rhino), Sumatran and Javan rhinos.
- **The Sumatran, Javan and Black rhinos are listed as 'critically endangered' by the IUCN -** there are thought to be fewer than 70 Javan rhinos and between 34-47 Sumatran rhinos left, which makes them truly under threat of extinction in the wild.
- **The Indian rhinos,** also known as the **greater one-horned rhino,** is the largest rhino species on Earth. They are native to the Indian subcontinent and are primarily found in India and Nepal. **African rhinos have a less armoured look and are more aggressive since they have two horns. So, statement 1 is correct.**
- Indian rhinos are strong swimmers and are considered the **best swimmers among all rhino species.** They can cross rivers without trouble and are often described as "semi-aquatic" due to their affinity for water. They frequently inhabit wetlands, swamps, and riverbanks, and are known to **dive underwater to feed on aquatic plants.** But their African relatives are very poor swimmers and **can drown in deep water. So, statement 2 is not correct.**

Therefore, option (a) is the correct answer.
Relevance: Wildlife experts and educationists have slammed the National Council of Educational Research and Training (NCERT) for mislabeling an African rhino as the Indian rhino in a class 4 mathematics textbook.

Q93.
Answer: d
Explanation:

- The Protection of Children from Sexual Offences Act, 2012 (POCSO) was enacted with the primary objective of protecting children from sexual offences and establishing a child-friendly mechanism for the trial of such offences. **One of the most significant changes brought by POCSO was the increase in the age of sexual consent from sixteen to eighteen years for both boys and girls. So, statement 1 is correct.**
- Section 19 of the POCSO states that anyone having "apprehension" that an offence under POCSO is likely to be committed, or knowledge that such an offence has been committed, "shall" provide such information to the Special Juvenile Police Unit (SJPU) or the police. The term "any person" also includes a child who may report an offence. This reporting provision becomes **binding due to Section 21 of the POCSO Act,** which prescribes punishment for **failing to report the commission of an offence under Section 19.** Imprisonment of six months to one year, or a fine, or both is prescribed under Section 21. **So, statement 2 is correct.**
- **POCSO is a gender-neutral law,** meaning it applies regardless of the **victim's gender.** As for the person who commits the crime - it does not matter **whether the person is male or female.** Both genders are equally punishable for offences under this law. **So, statement 3 is correct.**

Therefore, option (d) is the correct answer.
Relevance: The Supreme Court agreed to hear concerns flagged by senior advocate Indira Jaising that mandatory reporting of sexual activity under the Protection of Children from Sexual Offences (POCSO) is leading to the criminalisation of the young and poses a threat to the fundamental right to health of adolescent girls.

Q94.
Answer: b
Explanation:

- **The Central Drugs Standard Control Organisation (CDSCO)** under the Directorate General of Health Services, Ministry of Health & Family Welfare, Government of India, is the Central Drug Authority for discharging functions assigned to the Central Government under the **Drugs and Cosmetics Act 1940.**
- It has regulatory control over **the import of drugs, approval of new drugs and clinical trials,** meetings of the Drugs Consultative Committee (DCC) and the Drugs Technical Advisory Board (DTAB), and approval of certain licenses, **as the Central License Approving Authority, is exercised by the CDSCO. So, statement 1 is correct.**
- The essential medicines listed under the **National List of Essential Medicines (NLEM) are subject to price control as per the Drugs (Prices Control) Order, 2013** and the prices of 'Essential Medicines' shall be in accordance with the 'Ceiling Price' fixed by the **National Pharmaceutical Pricing Authority (NPPA)**. **So, statement 2 is not correct.**

Therefore, option (b) is the correct answer.
Relevance: The Drugs Controller General of India has been urged to lay down strict pathways for the newer antibiotics to curb antibiotic resistance.

Q95.
Answer: c
Explanation:

- **Bonobos share 98.7 percent of their genetic makeup with Humans,** hence are considered the closest cousins of Humans. They are about two-thirds the size of humans and are covered with dark hair. They are distinguished by their black faces, red lips, two or three webbed toes, a tail tuft, and parted long hair.
- Unlike their closest cousins, the common chimpanzees, they are not divided into subspecies. **They are found only in the Democratic Republic of Congo**. Their discontinuous range extends from the Lualaba River in the east to the Kasai/Sankuru rivers in the south and as far as the Congo River in the west and around Lake Tumba/Lake Mai-Ndombe.

Therefore, option (c) is the correct answer.
Relevance: Decades of observations in Congo lend support to the idea of a sisterhood in which female bonobos band together to assert their power.

Q96.
Answer: a
Explanation:

- **The River Cities Alliance (RCA) has been conceptualized to help river cities in India maintain this symbiotic relationship with their rivers**. Launched in November 2021 with 30 river cities, the RCA has now expanded to 145+ cities. The River Cities Alliance (RCA) has been launched with the objective to provide the member cities with a platform to discuss and exchange information on aspects that are vital for the sustainable management of urban rivers, sharing best practices and supporting innovation. The Alliance is open to all river cities of India. Any river city can join the Alliance at any time. **So, statement 1 is correct.**

- The RCA is currently being managed by the **National Mission for Clean Ganga (NMCG)** under the Ministry of Jal Shakti and the National Institute of Urban Affairs (NIUA) under the Ministry of Housing and Urban Affairs**. So, statement 2 is not correct.**

Therefore, option (a) is the correct answer.

Relevance: The National Mission for Clean Ganga (NMCG) has approved the Action Plan 2025 for the River Cities Alliance (RCA), focusing on integrating river-sensitive urban planning across India's expanding cities.

Q97.

Answer: c

Explanation:

- With the Prime Minister helming it, the Cabinet Committee on Security (CCS) has the ministers for Finance, Defence, Home Affairs, and External Affairs as its members. It is responsible for debates, discussions, and appointments/in the national security bodies.
- Apart from the Prime Minister, the Cabinet Committee on Economic Affairs comprises the **Defense Minister, the Home Minister, the Finance Minister, the External Affairs Minister, the Road Transport and Highways Minister, the Agriculture Minister, and the Heavy Industries and Steel Minister. So, points 1,2,4,6 and 7 are correct.**

Therefore, option (c) is the correct answer.

Relevance: Noting "the cross-border linkages of the terrorist attack" in Pahalgam on Tuesday (April 22) that led to the deaths of 26 people, the Cabinet Committee on Security (CCS) announced a series of extraordinary retaliatory measures against Pakistan.

Q98.

Answer: b

Explanation:

- **The mandatory upfront payment requirement for underground mining ventures has been completely waived by the Ministry of Coal**. This measure removes a significant financial barrier, encouraging broader participation from the private sector and facilitating faster project implementation. **So, statement 1 is not correct.**
- **To address the traditional challenges of high capital investment and longer gestation periods, the floor percentage of revenue share for underground coal mines has been reduced from 4% to 2% by the Ministry of Coal**. This targeted reduction offers substantial fiscal relief and enhances the financial viability of underground projects. **So, statement 2 is correct.**
- **An existing 50% rebate on performance security for underground coal blocks has been provided by the Coal Ministry, collectively lowering the entry threshold and facilitating smoother project implementation. So, statement 3 is correct.**

Therefore, option (b) is the correct answer.

Relevance: In a decisive step towards revitalizing India's coal sector, the Ministry of Coal has introduced a series of transformative policy measures aimed at promoting underground coal mining.

Q99.

Answer: a

Explanation:

- **A tonnage tax system is a special taxation policy for shipping companies where tax is not based on actual profits but on the size (tonnage) of the vessel**. It provides stable, predictable, and lower taxation for vessel owners, reducing their financial burden. **So, statement 1 is correct.**
- **Vide the Finance Bill, 2025, the said Scheme is proposed to be extended to the inland vessels operating in inland waterways in India. So, statement 2 is correct.**

Therefore, option (a) is the correct answer.

Relevance: The Government has also enacted the Inland Vessels Act, 20215, which replaced the Inland Vessels Act, 1916, and introduced a uniform regulatory framework for inland vessel navigation across the country.

Q100.

Answer: c

Explanation:

- **The GSTAT is the Appellate Authority established under the Central Goods and Services Tax Act, 2017**, to hear various appeals under the said Act and the respective State/Union Territories GST Acts against the orders of the first appellate authority. **So, statement 1 is correct.**
- In May 2024, the government appointed Justice (Retd) Sanjaya Kumar Mishra as the first President of the GST Appellate Tribunal (GSTAT). **Justice Mishra was a former Chief Justice of the Jharkhand High Court and was selected by a Search-cum-Selection Committee headed by the Chief Justice of India. So, statement 2 is not correct.**
- GSTAT consists of a principal bench and various state benches. As per the approval of the GST Council, the government has notified the Principal Bench, to be located in New Delhi, and 31 State benches at various locations across the country. **The Goods and Services Tax Appellate Tribunal (Procedure) Rules provide for mandatory e-filing of applications and conducting hearings in a hybrid mode. So, statement 3 is correct.**

Therefore, option (c) is the correct answer.

Relevance: The government has recently notified the Goods and Services Tax Appellate Tribunal (Procedure) Rules.

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